

Improved Bounds for the Grothendieck Constant

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September 10, 2026

Abstract

In this paper, we improve both the lower and upper bounds on the real Grothendieck constant:

$$1.77128 \leq K_G \leq 1.78052.$$

The previous best lower and upper bounds were $1.71359\dots$ and $1.78186\dots$, respectively, both due to Saha, Li, Xue, Chaudhuri, Klivans, Kothari, and Meka [SLX⁺26].

1 Introduction

Grothendieck’s inequality [Gro53], in the bilinear form given to it by Lindenstrauss and Pełczyński [LP68], asserts that there is a universal constant $K < \infty$ with the following property: for all positive integers m and n , every matrix $A \in \mathbb{R}^{m \times n}$, every real Hilbert space H , and all unit vectors $u_1, \dots, u_m, v_1, \dots, v_n \in H$,

$$\sum_{i=1}^m \sum_{j=1}^n A_{i,j} \langle u_i, v_j \rangle \leq K \cdot \max_{x \in \{-1,1\}^m, y \in \{-1,1\}^n} \sum_{i=1}^m \sum_{j=1}^n A_{i,j} x_i y_j. \quad (1)$$

The Grothendieck constant K_G is the infimum of the constants K for which Eq. (1) holds for every such choice. The right hand side of Eq. (1) is the value of an integer program, the left hand side becomes, after the supremum over the vectors is taken, the value of the natural semidefinite relaxation of that program, and K_G is by definition the largest ratio between the two. So K_G measures exactly how much is lost when a $\{-1,1\}$ bilinear optimization problem is relaxed to a semidefinite one.

More than seventy years after Grothendieck introduced it, the value of K_G is not known; Table 1 lists every improvement to the numerical bounds. Write $K_{\text{Kr}} := \pi/(2 \ln(1 + \sqrt{2})) = 1.78221\dots$ for Krivine’s bound and $K_{\text{DR}} := 1.67696\dots$ for the Davie–Reeds bound. Grothendieck [Gro53] proved $\pi/2 \leq K_G \leq \sinh(\pi/2)$, Rietz [Rie74] lowered the upper bound to 2.261, and Krivine [Kri79] lowered it to K_{Kr} by rounding the Hilbert space vectors with a random hyperplane after a preliminary transformation of them, a scheme that later became standard in approximation algorithms through the work of Goemans and Williamson [GW95]. Krivine also determined the two dimensional constant $K_G(2) = \sqrt{2}$, and he conjectured that $K_G = K_{\text{Kr}}$. In the other direction, a high dimensional Gaussian construction found independently by Davie [Dav84] and Reeds [Ree91] gives $K_G \geq K_{\text{DR}}$. Braverman, Makarychev, Makarychev and Naor [BMMN13] disproved Krivine’s conjecture by showing that $K_G < K_{\text{Kr}} - \varepsilon_0$ for some $\varepsilon_0 > 0$, but their argument produces no explicit value of ε_0 , and the Davie–Reeds bound went unimproved for more than four decades.

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Table 1: Every improvement to the numerical bounds on the Grothendieck constant K_G , in chronological order within each block. Here $K_{\text{Kr}} := \pi/(2\ln(1 + \sqrt{2}))$ is Krivine’s bound, $K_{\text{DR}} := 1.67696\dots$ is the Davie–Reeds bound, and $\varepsilon_0 > 0$ is the unspecified constant of [BMMN13]. The last row of each block is proved in this paper: K_L and K_U denote the exact bounds of Theorem 1.2 and Theorem 1.1, respectively. The bounds proved here are rounded outward to five decimal places, as in the abstract; other values are rounded to the nearest five-decimal value. A dash marks a bound with no closed form and a bound with no known numerical value.

Year	References	Lower bound	Value
1953	[Gro53]	$\pi/2$	1.57080
1984/1991	[Dav84, Ree91]	K_{DR}	1.67696
2026	[Hei26a]	$K_{\text{DR}} + 10^{-26}$	1.67696
2026	[JM26]	$K_{\text{DR}} + 10^{-12}$	1.67696
2026	[SLX+26]	$6\pi/11$	1.71360
2026	Theorem 1.2	K_L	1.77128
Year	References	Upper bound	Value
1953	[Gro53]	$\sinh(\pi/2)$	2.30130
1974	[Rie74]	—	2.26100
1979	[Kri79]	K_{Kr}	1.78221
2013	[BMMN13]	$K_{\text{Kr}} - \varepsilon_0$	—
2026	[Hei26b]	$K_{\text{Kr}} - 10^{-5}$	1.78220
2026	[LSX+26]	$K_{\text{Kr}} - 6.039 \cdot 10^{-5}$	1.78215
2026	[SLX+26]	$K_{\text{Kr}} - 3.4737 \cdot 10^{-4}$	1.78187
2026	Theorem 1.1	K_U	1.78052

Both sides moved in 2026. Heilman [Hei26a] and, independently, Jones and Malavolta [JM26] gave the first improvements over K_{DR} , by 10^{-26} and by 10^{-12} . Heilman [Hei26b] and, independently, Li, Saha, Xue, Chaudhuri, Klivans, Kothari and Meka [LSX+26] made ε_0 explicit, the latter reaching $\varepsilon_0 = 6.039 \cdot 10^{-5}$. Saha, Li, Xue, Chaudhuri, Klivans, Kothari and Meka [SLX+26] then improved both sides at once to

$$\frac{6\pi}{11} \leq K_G \leq K_{\text{Kr}} - 3.4737 \cdot 10^{-4}, \quad (2)$$

so that $1.71359 \leq K_G \leq 1.78187$. Their lower bound is the first one that is not read off an explicit gap instance; it comes instead from a limitation on asymptotically optimal Krivine type rounding schemes. The two sides of Eq. (2) agree to one decimal place, so the tenths digit of K_G is 7. The complex constant $K_G^{\mathbb{C}}$ is in a similar position: Haagerup [Haa87] proved $K_G^{\mathbb{C}} \leq 1.40491\dots$, and its exact value is open as well.

The interval in Eq. (2) is not only a question about tensor norms. Alon and Naor [AN06] turned Krivine’s rounding scheme into a polynomial time algorithm that approximates the cut norm of a matrix within the factor 0.56, and Khot and Naor [KN12] survey a range of combinatorial optimization problems whose best known approximation guarantees are Grothendieck type constants. Raghavendra and Steurer [RS09] gave an algorithm that approximates K_G to any prescribed additive accuracy in time depending only on that accuracy. In quantum information, Tsirelson [Tsi87] identified K_G as the supremal ratio between the entangled and classical biases of two-player XOR games, so any progress on Eq. (2) is progress on the known limits of nonlocality. Pisier [Pis12] surveys the subject as a whole.

The goal of this paper is to narrow the interval in Eq. (2).

1.1 Our results

We state our results as follows.

Theorem 1.1 (Upper bound, informal version of Theorem 2.6). *The Grothendieck constant satisfies*

$$K_G \leq \frac{625000\pi}{1102769} = 1.7805137871064\dots$$

Our lower bound is the following.

Theorem 1.2 (Lower bound, informal version of Theorem 3.20). *The Grothendieck constant satisfies*

$$K_G \geq \frac{107544546318225\pi + 1199235870000\sqrt{\pi/2}}{190089093929531 + 1199235870000\sqrt{\pi/2}} = 1.7712846852087007\dots$$

2 An improved upper bound

We modify the limiting rounding construction of [SLX⁺26] with a trigonometric threshold in two distinguished Gaussian coordinates and signed odd reservoirs. A sixth-order nuclear estimate controls the entire coefficient tail, and a complete finite-array norm check sharpens the integration-error propagation. The finite integration rules are specified in Appendix A.3. All terminating decimals denote exact rational numbers, and all Hermite indices are nonnegative integers.

Let $h_j = \text{He}_j / \sqrt{j!}$ be the orthonormal probabilists' Hermites. Define $u(x, y) = v(x) + v(y)$, where

$$\begin{aligned} v(x) = & 1.7696084086083774x - 4.493997202209279 \sin(x/2) \\ & + 0.3405759687395843 \sin x + 0.020267598042328753 \sin(3x/2). \end{aligned} \quad (3)$$

Put $\rho(t) = \sum_{j=1,3,\dots,21} r_j t^j$, with the following exact coefficients.

j	r_j	j	r_j
1	0.89890698293395284583943	13	-0.0000001454704750720801
3	-0.10013219150788148	15	-0.0000011182533010131454
5	0.0009168472487495326	17	0
7	0	19	-0.00000020485149584536287
9	-0.000027643327757993038	21	-0.0000002850103374914122
11	0.000014581396048726522		

Here $r_1 = 1 - \sum_{j \geq 3} |r_j| > 0$, so $\sum_j |r_j| = 1$ exactly. The reservoir variance is $|r_j|$ and its amplitude is $\sqrt{|r_j|}$; no further normalization is made. The positive linear coefficient r_1 supplies an independent Gaussian coordinate.

For independent standard Gaussians W, X, Y , define

$$\begin{aligned} A_{a,b,c} &= \mathbb{E}[h_a(W)h_b(X)h_c(Y) \text{sgn}(W + u(X, Y))], \\ H(t) &= \frac{\pi}{2} \sum_{a,b,c \geq 0} A_{a,b,c}^2 \rho(t)^a (-t)^{b+c} = \sum_{m \text{ odd}} b_m t^m. \end{aligned}$$

Oddness gives $A_{a,b,c} = 0$ when $a + b + c$ is even. There is no multinomial factor in this tensor Hermite basis. Parseval gives $\sum A_{a,b,c}^2 = 1$. Since the absolute coefficient sum of ρ is one, the

coefficient series for H is absolutely summable and defines a holomorphic function on the open unit disk. For $F(z) = \sum_j f_j z^j$, write $M_F(r) = \sum_j |f_j| r^j$.

An absolute coefficient bound for H yields a valid rounding scheme through its inverse.

Lemma 2.1 (Absolute inverse and finite transfer). *If $b_1 > 0$, $\Delta = \sum_{m \geq 3} |b_m|$ and $0 < \gamma < b_1 - \Delta$, then $M_{H^{-1}}(\gamma) < 1$ and $K_G \leq \pi/(2\gamma)$.*

Proof. Write $H = b_1 t + R_0$. In the zero-constant-term subspace of the Wiener algebra at radius γ , start with $U_0 = 0$ and iterate $U_{n+1} = (z - R_0(U_n))/b_1$. The unit ball maps into the ball of radius $(\gamma + \Delta)/b_1 < 1$. Each formal coefficient stabilizes because R_0 has order at least three. Fatou's lemma bounds the limiting Wiener norm; formal substitution and absolute convergence give an analytic inverse G with $M_G(\gamma) < 1$.

For a finite construction use independent Gaussian coordinates z, x, y and, separately for each odd $j \in \{3, 5, \dots, 21\}$ with $r_j \neq 0$, coordinates $Z_{j,1}, \dots, Z_{j,N}$. Define

$$f_N = \operatorname{sgn}\{\sqrt{r_1}z + u(x, y) + \sum_j \sqrt{|r_j|/N} \sum_{k=1}^N h_j(Z_{j,k})\},$$

$$g_N = \operatorname{sgn}\{\sqrt{r_1}z - u(x, y) + \sum_j \operatorname{sgn}(r_j) \sqrt{|r_j|/N} \sum_{k=1}^N h_j(Z_{j,k})\}.$$

At input correlation t , the joint central limit theorem gives a unit-variance reservoir pair with covariance $\rho(t)$. Negating both distinguished coordinates on the second side changes their correlations to $-t$ and turns $-u$ into u . The independent z coordinate makes every threshold-zero event null. Thus the normalized finite correlations H_N converge pointwise to H . Mehler and Cauchy-Schwarz give $\sum_m |[t^m]H_N| \leq \pi/2$; normal-family compactness gives local uniform convergence on the open unit disk.

Fix $0 < \gamma' < R' < \gamma$. Differentiating $H(G(z)) = z$ gives $H'(G(z))G'(z) = 1$. The holomorphic implicit-function theorem along the compact graph of G on $|z| \leq R'$ yields inverse branches G_N converging uniformly on a neighborhood of that disk. Uniqueness joins the local branches. Cauchy estimates give $M_{G_N - G}(\gamma') \rightarrow 0$, so $M_{G_N}(\gamma') < 1$ eventually. If $G_N(z) = \sum c_k z^k$, use the tensor maps

$$v \mapsto \bigoplus_k \sqrt{|c_k|(\gamma')^k} v^{\otimes k}, \quad w \mapsto \bigoplus_k \operatorname{sgn}(c_k) \sqrt{|c_k|(\gamma')^k} w^{\otimes k}.$$

Their norms are below one; pad the two families in mutually orthogonal directions. Every finite family spans a finite-dimensional Hilbert space. Gaussian rounding by f_N, g_N then has expected product $2\gamma' \langle v, w \rangle / \pi$. The finite Grothendieck inequality follows with constant $\pi/(2\gamma')$. Let $\gamma' \uparrow \gamma$. $\square$

2.1 Trace-class Gaussian kernels

All spatial spaces below use standard Gaussian measure. For real $|r| < 1$ and $x, y \in \mathbb{R}^2$ put

$$C_r(x, y) = S_r(u(x), u(y)), \quad S_r(v, w) = \frac{\pi}{2} \mathbb{E}[\operatorname{sgn}(W + v) \operatorname{sgn}(W' + w)], \quad \mathbb{E}[WW'] = r.$$

We use the same notation for the associated integral operator, and write $\mathcal{C}(t) = C_{\rho(t)}$. The symbol $\|\cdot\|_1$ denotes the nuclear norm of an operator, not a scalar Wiener norm.

To control large temporal indices, we first bound the kernel on a complex strip.

Lemma 2.2 (Unweighted trace bound). *The operators C_r extend holomorphically in nuclear norm to $|\Re r| < 1$, and*

$$\|C_r\|_1 \leq \frac{\pi}{2} \sqrt{\frac{|1 - r^2|}{1 - (\Re r)^2}}.$$

The same estimate holds for any measurable, finite-almost-everywhere real threshold on a probability space with separable L^2 .

Proof. Put $f_w(x) = \text{sgn}(w + u(x))$. Each f_w has L^2 norm at most one. With the square-root branch positive at zero, the complex Gaussian density is

$$\varphi_r(w, z) = \frac{1}{2\pi\sqrt{1 - r^2}} \exp\left(-\frac{w^2 - 2rwz + z^2}{2(1 - r^2)}\right).$$

It gives the rank-one representation

$$C_r = \frac{\pi}{2} \int_{\mathbb{R}^2} \varphi_r(w, z) f_w \otimes f_z \, dw \, dz.$$

Here the bilinear rank-one kernel acts by

$$(f \otimes g)h(x) = f(x) \int g(y)h(y) \, d\gamma(y)$$

on the complexified space, and has nuclear norm $\|f\|_2\|g\|_2$. Writing $a = \Re r$, the real inverse-covariance eigenvalues are $(1 + a)/|1 + r|^2$ and $(1 - a)/|1 - r|^2$. Consequently

$$\int_{\mathbb{R}^2} |\varphi_r(w, z)| \, dw \, dz = \sqrt{\frac{|1 - r^2|}{1 - a^2}}.$$

The strongly measurable rank-one integrand is therefore integrable in nuclear norm. On compact subsets of the strip, this Gaussian domination and the corresponding bounds after parameter differentiation are locally uniform. The Bochner integral proves both the estimate and nuclear-norm holomorphy. $\square$

For $p \geq 1$ define the full derivative tensor

$$D_p f = \left(\sqrt{\binom{p}{\alpha}} \partial^\alpha f \right)_{|\alpha|=p}.$$

This symmetric-tensor notation retains all ordered directions. Generate the Bell polynomials by

$$B_{0,0} = 1, \quad B_{\alpha+e_j,i} = \partial_j B_{\alpha,i} + (\partial_j u) B_{\alpha,i-1},$$

with absent entries zero. Let $\mathcal{K}_{p,r}$ be the operator on the vector-valued Gaussian space with block kernel

$$\left(\sqrt{\binom{p}{\alpha} \binom{p}{\delta}} \partial_x^\alpha \partial_y^\delta C_r(x, y) \right)_{|\alpha|=|\delta|=p}.$$

Define the real Gram matrix

$$\mu_{ij}^{(p)} = \sum_{|\alpha|=p} \binom{p}{\alpha} \mathbb{E}[B_{\alpha,i}(X) B_{\alpha,j}(X)] \quad (1 \leq i, j \leq p).$$

Spatial derivatives supply the Hermite-degree weights needed to control large spatial indices.

Lemma 2.3 (Weighted nuclear estimate). *For the threshold above, $\mathcal{K}_{p,r}$ is holomorphic in nuclear norm on $|\Re r| < 1$. With $z = (1 - (\Re r)^2)^{-1}$,*

$$\|\mathcal{K}_{p,r}\|_1 \leq \sqrt{z} \sum_{\substack{1 \leq i,j \leq p \\ i+j \text{ even}}} (-1)^{(i-j)/2} (i+j-3)!! \mu_{ij}^{(p)} z^{(i+j-2)/2},$$

where $(-1)!! = 1$.

Proof. Gaussian differentiation gives

$$\partial_v \partial_w S_r(v, w) = g_r(v, w) = \frac{1}{\sqrt{1-r^2}} \exp\left(-\frac{v^2 - 2rvw + w^2}{2(1-r^2)}\right).$$

On the strip this has the absolutely convergent Fourier representation

$$g_r(v, w) = \frac{1}{2\pi} \int_{\mathbb{R}^2} e^{-(\xi^2 + 2r\xi\eta + \eta^2)/2} e^{i\xi v} e^{i\eta w} d\xi d\eta.$$

The real Gaussian Fourier identity extends to the strip by dominated holomorphy. Define

$$\psi_\xi(x) = \left(\sqrt{\binom{p}{\alpha}} e^{i\xi u(x)} \sum_{j=1}^p (i\xi)^{j-1} B_{\alpha,j}(x) \right)_{|\alpha|=p}.$$

The two chain rules give

$$\mathcal{K}_{p,r} = \frac{1}{2\pi} \int_{\mathbb{R}^2} e^{-(\xi^2 + 2r\xi\eta + \eta^2)/2} \psi_\xi \otimes \psi_\eta d\xi d\eta.$$

The absolute Fourier weight is symmetric in ξ, η . Applying $2\|\psi_\xi\|_2 \|\psi_\eta\|_2 \leq \|\psi_\xi\|_2^2 + \|\psi_\eta\|_2^2$ and integrating one variable yields

$$\|\mathcal{K}_{p,r}\|_1 \leq \sqrt{z} \mathbb{E}_{\xi \sim \mathcal{N}(0,z)} \|\psi_\xi\|_2^2.$$

Since u is real, $|e^{i\xi u}| = 1$. Expanding the square, odd Gaussian moments vanish. For $i+j$ even the phase is $i^{i-1}(-i)^{j-1} = (-1)^{(i-j)/2}$ and the moment is $(i+j-3)!! z^{(i+j-2)/2}$, giving the claimed bound. All positive-order derivatives of the trigonometric threshold are bounded. Thus the Bell coefficients are bounded, and locally uniform Gaussian domination justifies all differentiations and the nuclear-norm holomorphy. $\square$

For $p = 6$ write the right side as $\sqrt{z} \sum_{j=0}^5 d_j z^j$, where

$$d_j = (2j-1)!! \sum_{\substack{1 \leq i,l \leq 6 \\ i+l=2j+2}} (-1)^{(i-l)/2} \mu_{il}^{(6)}.$$

Use the exact caps

$$(\bar{d}_0, \dots, \bar{d}_5) = (0.244845, 6.685184, 296.722886, \\ 4208.318470, 27527.734330, 70846.926235)$$

and radii

$$R = \frac{107}{100}, \quad r_* = \frac{5}{4}, \quad B_6 = 28220000.$$

The chosen radii give separate bounds for the temporal tail and the sixth-order weighted trace.

Lemma 2.4 (Complete circle bounds). *On $|t| \leq r_*$ one has $|\Re \rho(t)| < 39/40$, $|\rho(t)| < 53/40$, and $\|\mathcal{C}(t)\|_1 < 12$. Moreover*

$$\frac{1}{2\pi} \int_0^{2\pi} \|\mathcal{K}_{6,\rho(Re^{i\theta})}\|_1 d\theta < B_6.$$

Finite interval proof. The exact Fourier calculation in Appendix A.3 proves $0 < d_j < \bar{d}_j$. Partition $[0, \pi/2]$ into 16384 equal closed arcs. Whole-arc evaluation of ρ at radius $5/4$ gives

$$|\Re \rho(t)| < 0.971031784, \quad |\rho(t)| < 1.321639198.$$

The harmonic maximum principle and the maximum-modulus principle give the disk bounds. Lemma 2.2 then gives

$$\|\mathcal{C}(t)\|_1^2 \leq \left(\frac{\pi}{2}\right)^2 \frac{1 + (53/40)^2}{1 - (39/40)^2} < 137.706 < 144.$$

At radius $107/100$, let X_j enclose $|\Re \rho(t)|$ on the entire j th arc. All $\bar{d}_l$ are positive, so

$$z_j = (1 - X_j^2)^{-1}, \quad \sqrt{z_j} \sum_{l=0}^5 \bar{d}_l z_j^l$$

bounds the nuclear norm there. The mean enclosing upper endpoint is less than $28209773.802 < 28220000$. Conjugation and oddness of ρ give the identical majorant on the other three quarters; the factor four cancels in the normalized angular mean. Every arc passes the printed tests. Strictness and compactness give the neighborhoods of the closed disks needed for Cauchy's formula. $\square$

2.2 A weighted spatial Hermite trace

For $n = (n_1, n_2) \in \mathbb{N}^2$, write $h_n(x) = h_{n_1}(x_1)h_{n_2}(x_2)$ and $|n| = n_1 + n_2$. Put

$$a_{n,m,k} = [t^k] \langle \mathcal{C}(t), h_n \otimes h_m \rangle, \quad (s)_p = s(s-1) \cdots (s-p+1).$$

Hermite differentiation and the falling-factorial identity give

$$\langle D_p h_n, D_p h_m \rangle = \mathbf{1}_{\{n=m\}} (|n|)_p, \quad \sum_{|\alpha|=p} \binom{p}{\alpha} (n)_\alpha = (|n|)_p.$$

Hence $e_n = D_p h_n / \sqrt{(|n|)_p}$, $|n| \geq p$, is an orthonormal family in the vector-valued Gaussian space. It need not be complete. Gaussian integration by parts in both variables gives

$$\langle \mathcal{K}_{p,\rho(t)} e_n, e_n \rangle = (|n|)_p \langle \mathcal{C}(t), h_n \otimes h_n \rangle.$$

Indeed, the adjoint derivative applied to the α component of $D_p h_n$ gives $(n)_\alpha h_n$, and summing the binomial weights gives $(|n|)_p$. Bounded threshold derivatives and Gaussian decay justify integration by parts. The identity holds first for real t near zero and then throughout the analytic disk.

For a nuclear operator T and any orthonormal family,

$$\sum_n |\langle T e_n, e_n \rangle| \leq \|T\|_1.$$

For a rank-one decomposition this follows from Cauchy–Schwarz and Bessel’s inequality; take the infimum over such decompositions. Nuclear-norm Cauchy estimates and Lemma 2.4 therefore give

$$\sum_{|n| \geq 6} (|n|)_6 |a_{n,n,k}| \leq B_6 R^{-k}, \quad \sum_n |a_{n,n,k}| \leq 12 r_*^{-k}.$$

Every multiindex occurs in these sums. In particular the $s + 1$ indices of spatial degree s have not been replaced by one index; their entire sum is controlled by the nuclear norm.

Mehler’s identity gives

$$b_m = \sum_{|n|+k=m} (-1)^{|n|} a_{n,n,k}.$$

One may verify this first for small real t by conditioning on the reservoir coordinate and then equating Taylor coefficients. Each fixed coefficient has finitely many indices. The unweighted estimate above also gives absolute summability over all diagonal and temporal indices.

We split the temporal index to combine the weighted and unweighted estimates for the omitted coefficients.

Lemma 2.5 (Weighted infinite tail). *For this construction,*

$$\sum_{m \geq 303, m \text{ odd}} |b_m| < T_* := 0.00000085.$$

Proof. Split the source temporal index at $k = 107$. For $0 \leq k \leq 107$, an omitted output term has $|n| \geq 303 - k \geq 196$. Since $(s)_6$ increases for integers $s \geq 6$, its full spatial sum is at most $B_6 R^{-k} / (303 - k)_6$. For $k \geq 108$, use the unweighted estimate on every spatial index, including the low ones. Thus

$$\sum_{m \geq 303, m \text{ odd}} |b_m| \leq 28220000 \sum_{k=0}^{107} \frac{(100/107)^k}{(303 - k)_6} + 60(4/5)^{108} < \frac{85}{10^8}.$$

The last comparison is rational and passes after clearing positive denominators. The two terms are respectively less than 0.000000843909691 and 0.000000002050548. The split covers every omitted output coefficient, every spatial degree and the complete infinite temporal exterior. No continuation of the traced function H beyond the unit disk is used. $\square$

We combine the finite coefficient certificate with the infinite-tail estimate to complete the upper bound.

Theorem 2.6 (Upper bound, formal version of Theorem 1.1).

$$K_G \leq \frac{625000\pi}{1102769} = 1.7805137871064\dots < 1.78051379. \quad (4)$$

Proof. Put $\gamma_U = 1102769/1250000 = 0.8822152$. Lemmas A.2 and 2.5 give

$$b_1 - \sum_{m \geq 3} |b_m| > \ell_* - T_* = \gamma_U.$$

The last equality is the exact rational identity

$$0.88221605 - 0.00000085 = 0.8822152.$$

Lemma 2.1 gives $K_G \leq \pi / (2\gamma_U)$. $\square$

3 A certified lower bound

We prove the lower bound directly from one family of fiber inequalities. Oblique clipping and phase-compatible matrix comparisons reduce the problem to a four-variable dual. Retained matrix remainders and disagreement-mass bounds then give a single scalar cover. The endpoint is handled by a concavity argument, and an antipodal finite-matrix transfer retains the odd-projection term.

Let P_j be orthogonal projection onto total Hermite degree j in standard Gaussian L^2 , and put

$$\nu = \sqrt{2/\pi}, \quad P_{\text{odd}}f(x) = \frac{f(x) - f(-x)}{2}, \quad b_j = \frac{\pi}{2} \langle P_j f, P_j g \rangle.$$

For $\beta > 0$ and $\eta \geq 0$, write

$$\mathcal{B}_{\beta,\eta}(f, g) = \beta b_1 - b_3 + \frac{\eta}{\nu} \langle P_{\text{odd}}f, P_{\text{odd}}g \rangle, \quad \mathcal{A}_{\beta,\eta} = \mathcal{B}_{\beta,\eta} - \eta/\nu.$$

The matrix reference parameters are

$$\alpha_0 = 1.433929, \quad \kappa_0 = .0079952, \quad C_0 = \alpha_0 - 1/6.$$

They specify the fiber matrices and costs; no affine inequality at these reference parameters is assumed. The final parameters and bound are given in Theorem 3.20.

3.1 Agreement, phase information and weighted clipping

Initially let f, g be odd modulo null sets. Put

$$h = (f + g)/2, \quad k = (f - g)/2, \quad x = \|P_1 h\|_2/\nu.$$

Then h, k are ternary, $hk = 0$ and $h^2 + k^2 = 1$. Choose the first Gaussian coordinate S in the direction of $P_1 h$, with any direction if it is zero, and denote the transverse coordinates by Y . Thus $\mathbb{E}[Sh] = \nu x$ and $0 \leq x \leq 1$, since $|h| \leq 1$ and $\mathbb{E}|S| = \nu$. Define

$$q_1 = \|P_1 k\|_2^2/\nu^2, \quad p_j = \|P_j h\|_2^2/\nu^2, \quad z_j = \|P_j k\|_2^2/\nu^2 \quad (j = 3, 5).$$

Bilinearity gives $b_1 = x^2 - q_1$ and $b_3 = p_3 - z_3$. Set

$$c = -\mathbb{E}[(S^3 - 3S)h]/\nu, \quad p_3 \geq c^2/6,$$

where the last inequality retains the normalized third Hermite in S . The phase constants are

$$a_0 = 1.24607769, \quad \rho = 0.941845822, \quad \sigma = 0.82340390, \quad c_0 = 0.79327770.$$

All terminating decimals in this section denote exact rational numbers. Set $y = \mathbb{E}[|S|k^2]/\nu$, $z = \mathbb{E}[|S|^3 k^2]/\nu$ and $m = \mathbb{E}[k^2]$. The support identities imply $0 \leq y \leq 1 - x$ and $0 \leq z \leq 2 - 3x + c$.

A seven-copy Gaussian comparison couples the first three odd Hermite levels.

Lemma 3.1. *The agreement and disagreement energies satisfy*

$$\rho(x^2 + q_1) + p_3 + z_3 + \sigma(p_5 + z_5) \leq a_0. \quad (5)$$

Every odd measurable F with $|F| \leq a$ satisfies

$$\rho\|P_1 F\|_2^2 + \|P_3 F\|_2^2 \leq c_0 a^2.$$

Proof. For an odd sign b , take independent Gaussian vectors X, Y and put $X_j = X \cos(j\pi/7) + Y \sin(j\pi/7)$ and $s_j = b(X_j)$. Then $s_{j+7} = -s_j$ and

$$\sum_{j=0}^6 s_j(s_{j+1} - s_{j+3}) \leq 4.$$

Indeed, put $e_j = s_j s_{j+1}$ with periodic indices. Their product is -1 . If n counts negative e_j and m counts negative $e_j e_{j+1} e_{j+2}$, the sum is $2(m - n)$ and both counts are odd. If $n = 1$, then $m = 3$; if $n \geq 5$, then $m \leq n + 2$. For $n = 3$, the only bad case is $m = 7$, which forces $e_j = e_{j+3}$ for all j , hence all e_j equal, a contradiction.

With $u = \cos(\pi/7)$ and $v = \sin(\pi/14)$, Mehler's identity gives

$$7 \sum_{j \geq 1, j \text{ odd}} (u^j - v^j) \|P_j b\|_2^2 \leq 4.$$

The weights are positive. Retaining levels one, three and five gives the exact normalized coefficients

$$\rho_* = (u^2 + uv + v^2)^{-1}, \quad \sigma_* = (u^5 - v^5)/(u^3 - v^3), \quad a_* = 2\pi/(7(u^3 - v^3)).$$

The rational Taylor comparisons in Appendix A.2 give $\rho_* > \rho$, $\sigma_* > \sigma$, $a_* < a_0$ and $2a_0 < c_0\pi$. Apply the sign inequality to f, g , average and divide by ν^2 , using $\|P_j h\|_2^2 + \|P_j k\|_2^2 = (\|P_j f\|_2^2 + \|P_j g\|_2^2)/2$. This proves Eq. (5). For one odd sign the same estimate gives $\rho \|P_1 b\|_2^2 + \|P_3 b\|_2^2 \leq 2a_0/\pi < c_0$. For $a > 0$, choose an almost-everywhere odd sign ς and let U be uniform on $[-1, 1]$. The functions $\varsigma \operatorname{sgn}(\varsigma F/a - U)$ are odd signs averaging to F/a . Convexity gives the bounded-function estimate. The case $a = 0$ is immediate. $\square$

Writing $\mathcal{B} = \alpha_0 b_1 - b_3$, we obtain

$$\mathcal{B} \leq a_0 + (\alpha_0 - \rho)x^2 - (\alpha_0 + \rho)q_1 - 2p_3 - \sigma(p_5 + z_5). \quad (6)$$

Write $[t]_+ = \max(t, 0)$ and $\operatorname{clip}_a(t) = \max(-a, \min(t, a))$.

Clipping extends the bounded-function estimate to odd square-integrable functions.

Lemma 3.2 (Weighted clipping). *For odd square-integrable F and $a, \epsilon > 0$,*

$$\rho \|P_1 F\|_2^2 + \|P_3 F\|_2^2 \leq (1 + \epsilon)c_0 a^2 + (1 + \epsilon^{-1})\mathbb{E}[(|F| - a)_+^2].$$

Proof. The operator $A = \sqrt{\rho}P_1 + P_3$ is a contraction. Apply $\|u + v\|_2^2 \leq (1 + \epsilon)\|u\|_2^2 + (1 + \epsilon^{-1})\|v\|_2^2$ to $u = A \operatorname{clip}_a F$ and $v = A(F - \operatorname{clip}_a F)$. Lemma 3.1 bounds the first term, and contraction bounds the second by the clipping remainder. $\square$

3.2 Oblique matrix comparisons and transverse accounting

For a ternary function w on the line use raw normalized moments

$$m_j(w) = \mathbb{E}[\operatorname{He}_j(Z)w(Z)]/\nu, \quad m_e = (m_0, m_2)^\top, \quad m_o = (m_1, m_3)^\top.$$

In particular $\operatorname{He}_2 = z^2 - 1$ and $\operatorname{He}_3 = z^3 - 3z$ are not divided by factorial square roots. The factors $1/2$ and $1/6$ below account for this convention. Choose $0 < \theta \leq 1$, $r > 0$, $\zeta \geq 0$, $0 < \lambda \leq 1$, and $s_o \in \mathbb{R}$. Put

$$\beta = \frac{\alpha_0 + (1 - \theta)\rho}{\theta}, \quad \eta = \frac{(1 - \theta)\sigma}{\theta}, \quad \delta = 1 - \lambda, \quad w_o = (1, s_o)^\top.$$

Define

$$A_e = \text{diag}(-\beta, 1/2) - \delta \rho w_o w_o^\top, \quad B_e = \text{diag}(1, -\eta/2) - \delta w_o w_o^\top.$$

Let $M_e = B_e - A_e$, $v_r = (2r, 1)^\top$, and

$$d_r = v_r^\top \text{adj}(M_e) v_r, \quad t_e = -\det(M_e)/d_r, \quad Q_e = B_e + t_e v_r v_r^\top.$$

For every selected row $d_r, t_e > 0$, $Q_e \succ 0$, and

$$Q_e \succeq A_e, \quad Q_e \succeq B_e. \quad (7)$$

The second difference is a positive rank-one matrix. For the first, $\det(M_e + t_e v_r v_r^\top) = \det(M_e) + t_e d_r = 0$; its diagonals are nonnegative. These signs and the positivity of Q_e are checked by exact rational arithmetic. Explicitly,

$$\begin{aligned} (M_e)_{00} &= \beta + 1 - \delta(1 - \rho), & (M_e)_{01} &= -\delta(1 - \rho)s_o, & (M_e)_{11} &= -(1 + \eta)/2 - \delta(1 - \rho)s_o^2, \\ d_r &= (M_e)_{00} - 4r(M_e)_{01} + 4r^2(M_e)_{11}, \\ q_{00} &= 1 - \delta + 4r^2 t_e, & q_{01} &= -\delta s_o + 2r t_e, & q_{11} &= -\eta/2 - \delta s_o^2 + t_e, & Q_e &= (q_{ij})_{i,j=0,1}. \end{aligned}$$

For the odd block put

$$v_o = \frac{(\beta + 1)(1 + \eta)}{\beta + 1 - \zeta^2(1 + \eta)/6}, \quad Q_o = \begin{pmatrix} 1 + \zeta^2 v_o/6 & \zeta v_o/6 \\ \zeta v_o/6 & (v_o - \eta)/6 \end{pmatrix}.$$

Its denominator is positive, $Q_o \succ 0$, and

$$Q_o \succeq \text{diag}(-\beta, 1/6), \quad Q_o \succeq \text{diag}(1, -\eta/6). \quad (8)$$

Both difference determinants vanish; the diagonal and definiteness checks are rational. Set $W(w) = m_e^\top Q_e m_e + m_o^\top Q_o m_o$. For $\delta > 0$, choose $a, \epsilon > 0$ and let

$$\gamma = \delta(1 + \epsilon^{-1}), \quad J = \delta(1 + \epsilon)c_0 a^2.$$

For the ordinary row $\delta = 0$, take $s_o = 0$ and $\gamma = J = 0$; $a = 0, \epsilon = 1$ are only tabular conventions.

The matrix comparisons lift a one-dimensional fiber estimate to the full Gaussian space.

Lemma 3.3. *Suppose every ternary w satisfies*

$$W(w) + \gamma[|m_0(w) + s_o m_2(w)| - a]_+^2 \leq \mathbb{E}[\omega_\theta(Z)w(Z)^2]/\nu + I, \quad (9)$$

where $\omega_\theta(z) = 2\kappa_0/\theta + L|z| + R|z|^3$, $L > 0$, $R \geq 0$. Then

$$z_3 - \beta q_1 - \eta z_5 \leq \mathbb{E}[\omega_\theta(S)k(S, Y)^2]/\nu + I + J. \quad (10)$$

Proof. Let $K_j(Y) = \mathbb{E}_S[\text{He}_j(S)k(S, Y)]$ and let Π_ℓ project onto transverse degree ℓ . Raw-moment orthogonality gives

$$\|P_m k\|_2^2 = \sum_{j=0}^m \|\Pi_{m-j} K_j\|_2^2 / j!.$$

At transverse degree zero use the first odd comparison for (K_1, K_3) , discarding the nonpositive target term involving K_5 . At degree one use $Q_e \succeq A_e$ on (K_0, K_2) , leaving $\delta \rho \|\Pi_1(K_0 + s_o K_2)\|_2^2$; discard the nonpositive K_4 term. At degree two use the second odd comparison. At degree three use $Q_e \succeq B_e$, leaving $\delta \|\Pi_3(K_0 + s_o K_2)\|_2^2$. The degree-four and degree-five target terms are nonpositive, and all unused components of W are nonnegative. Divide by ν^2 and sum:

$$z_3 - \beta q_1 - \eta z_5 \leq \mathbb{E}_Y W(k_Y) + \delta(\rho \|\Pi_1 F\|_2^2 + \|\Pi_3 F\|_2^2), \quad F = (K_0 + s_o K_2)/\nu.$$

Joint oddness of k and evenness of He_2 make F odd in Y . Weighted clipping bounds the last term by $J + \gamma \mathbb{E}_Y[(|F| - a)_+^2]$. Average Eq. (9) over Y . In zero transverse dimension $K_0 = K_2 = 0$ by oddness, so the argument also applies there. Ordinary rows have no clipping term. $\square$

3.3 The four-variable nonlinear criterion

Let $D_e = \det Q_e > 0$ and put

$$e = q_{11}/D_e, \quad p = (q_{11} - q_{01})/D_e, \quad q = (q_{00} - 2q_{01} + q_{11})/D_e,$$

$$h_A = (q_{11} - s_o q_{01})/D_e, \quad h_B = (q_{11} - q_{01} + s_o(q_{00} - q_{01}))/D_e.$$

For $\delta > 0$ define

$$d_0 = \frac{q_{11} - 2s_o q_{01} + s_o^2 q_{00}}{D_e} + \gamma^{-1}, \quad H_o(A, B) = \frac{[|h_A A + h_B B| - a]_+^2}{d_0}.$$

For an ordinary row set $H_o = 0$ and omit all γ^{-1} expressions.

Quadratic duality absorbs the clipping term into the even variables, keeping the final criterion four-dimensional.

Lemma 3.4 (Elimination of the clipping dual variable). *For $m \in \mathbb{R}^2$, $Q_e \succ 0$ and $\gamma > 0$,*

$$m^\top Q_e m + \gamma[|w_o^\top m| - a]_+^2 = \sup_b \{2b^\top m - b^\top Q_e^{-1} b + [|w_o^\top Q_e^{-1} b| - a]_+^2 / d_0\}.$$

Proof. The penalty has the dual representation $\gamma[|t| - a]_+^2 = \sup_u (2ut - u^2/\gamma - 2a|u|)$. Combine this with quadratic duality and put $b = b_0 + uw_o$. For fixed b the remaining supremum is

$$-b^\top Q_e^{-1} b + \sup_u (2uw_o^\top Q_e^{-1} b - d_0 u^2 - 2a|u|).$$

Its maximizer is $\text{sgn}(w_o^\top Q_e^{-1} b)[|w_o^\top Q_e^{-1} b| - a]_+ / d_0$, which proves the formula. $\square$

For $b = (A+B, B)^\top$, the quadratic penalty is $eA^2 + 2pAB + qB^2$ and the added term is $H_o(A, B)$. For the odd block, on $z > 0$ put

$$v_0(z) = (w(z) - w(-z))/2, \quad d\mu(z) = ze^{-z^2/2} dz, \quad \phi(z) = (z^2 - 2)/2.$$

The functions $1, \phi$ are orthonormal in $L^2(\mu)$. If $m = \int v_0 d\mu$ and $n = \int \phi v_0 d\mu$, then $m_1 = m$, $m_3 = 2n - m$ and $|m|, |n| \leq 1$ by Cauchy-Schwarz. The odd form is $(m, n)G(m, n)^\top$, where

$$G = \begin{pmatrix} g_a & g_b \\ g_b & g_c \end{pmatrix}, \quad g_a = 1 - \eta/6 + v_o(\zeta - 1)^2/6, \quad g_b = (\eta + (\zeta - 1)v_o)/3, \quad g_c = 2(v_o - \eta)/3.$$

Write $G^{-1} = \begin{pmatrix} o_a & o_b \\ o_b & o_c \end{pmatrix}$ and set

$$m_* = g_a + |g_b|, \quad n_* = g_c + |g_b|, \quad |M| \leq m_*, \quad |N| \leq n_*. \quad (11)$$

The unrestricted odd dual maximizer is $G(m, n)^\top$, so restriction to this rectangle loses nothing. Define

$$P_{\epsilon_1, \epsilon_2}(z) = \epsilon_1 A - \kappa_0/\theta + [\epsilon_2(M - N) - L/2]z + \epsilon_1 B z^2 + (\epsilon_2 N - R)z^3/2, \\ \mathcal{F}_{\kappa_0}(A, B, M, N) = \sum_{\epsilon_1, \epsilon_2 \in \{-1, 1\}} \int_0^\infty [P_{\epsilon_1, \epsilon_2}(z)]_+ e^{-z^2/2} dz. \quad (12)$$

To deduce Eq. (9) from the theorem below, use even and odd quadratic duality. Put $P = A + Bz^2$ and $O = z(M - N) + Nz^3/2$. Since $1/(\nu\sqrt{2\pi}) = 1/2$, the pointwise expression after subtracting the normalized cost, before the quadratic penalties, is

$$w(z)(P + O) + w(-z)(P - O) - \frac{\omega_\theta(z)}{2}(w(z)^2 + w(-z)^2).$$

The two ternary choices maximize independently. Their maximum is $[|P + O| - \omega_\theta/2]_+ + [|P - O| - \omega_\theta/2]_+$. Nonnegativity of the cost makes this exactly the four-positive-part sum in Eq. (12). Integrate, apply the theorem, and take the dual suprema. In particular the odd cross term and the constant cost are both retained.

Theorem 3.5. *For every row specified in Section 3.5, all real A, B , and the entire rectangle in Eq. (11),*

$$\mathcal{F}_{\kappa_0} + H_o \leq eA^2 + 2pAB + qB^2 + o_a M^2 + 2o_b MN + o_c N^2 + I. \quad (13)$$

Proof. For each of the fifty-three rows, evaluate the deterministic procedure in Section 3.6 with 180 significant bits and root grid $2^{-80}\mathbb{Z}$, using the binary arithmetic specified in Appendix A.2. It exhausts its subdivision stack; the maximum depth is 69, below the failure limit 110. Each row has twelve exact-zero leaves, and every remaining accepted leaf has a strictly negative full-box upper bound. All rational matrix comparisons and exterior tests pass. Replays at 220/96 and with the explicit Taylor–Mills primitives also accept the covers. These replays check reproducibility; the error bounds are the enclosing formulas in Appendix A.2, not agreement between runs.

Lemma 3.8 or Eq. (17) proves the inequality on every accepted full box. Complete traversal covers the compact domain, the analytic exterior covers its complement, and symmetry covers the other sign halves. Thus the assertion uses the printed generating and acceptance rules together with the analytic estimates in Section 3.6. $\square$

3.4 Scalar coupling and support-aware closure

Because $h^2 \geq \text{sgn}(S)h$,

$$\mathbb{E}[|S|(L + RS^2)k^2]/\nu \leq L + 2R - (L + 3R)x + Rc.$$

Here $\mathbb{E}|S|^3 = 2\nu$ and $\mathbb{E}[S^3h]/\nu = 3x - c$.

Mixing the fiber and phase inequalities reduces the bound to the scalar variable x .

Lemma 3.6. *Every certified row gives*

$$\begin{aligned} \mathcal{B}_{\alpha_0, \kappa_0} \leq Q_\theta(x) + \kappa_0/\nu, \quad Q_\theta(x) = & [\alpha_0 - (1 - \theta)\rho]x^2 + (1 - \theta)a_0 \\ & + \theta[L + 2R - (L + 3R)x + I + J] + \frac{3\theta^2 R^2}{2(2 - \theta)}. \end{aligned} \quad (14)$$

This holds also for $\theta = 1$ and for all $0 \leq x \leq 1$.

Proof. Insert Eq. (10) in $\mathcal{B} = \alpha_0(x^2 - q_1) - p_3 + z_3$ and mix it with the phase bound, with weights θ and $1 - \theta$. The q_1, z_5 terms cancel since

$$\theta\beta - \alpha_0 - (1 - \theta)\rho = 0, \quad \theta\eta - (1 - \theta)\sigma = 0.$$

Discard the nonpositive agreement fifth-degree term. Since the signs are odd, $\langle f, g \rangle = 1 - 2\mathbb{E}k^2$. The constant fiber cost contributes $2\kappa_0\mathbb{E}k^2/\nu$, exactly canceled by the negative part of the odd-projection term. Finally use $p_3 \geq c^2/6$ and $\theta Rc - (2 - \theta)c^2/6 \leq 3\theta^2 R^2/(2(2 - \theta))$. $\square$

For pure rows retain $y = \mathbb{E}[|S|k^2]/\nu$, so $0 \leq y \leq 1 - x$.

Retaining this support constraint sharpens the scalar bound near $x = 1$.

Lemma 3.7. *For $0 < x \leq 1$, put*

$$P = (1 - y + x)/2, \quad Q = (1 + y + x)/2, \quad D(x, y) = x + 2(P \log P + Q \log Q), \quad D_+(x) = x + 2x \log x.$$

Then $c \geq D(x, y)$. For a pure row with $x \geq 1/2$, $L \geq 1/3$ and $R \geq 0$,

$$\mathcal{B}_{\alpha_0, \kappa_0} \leq Q_1(x) - [D_+(x) - 3R]_+^2/6 + \kappa_0/\nu. \quad (15)$$

Proof. For $s > 0$ put $p(s) = \Pr_Y[h(s, Y) = 1]$ and $n(s) = \Pr_Y[h(s, Y) = -1]$. Under $d\mu(s) = se^{-s^2/2}ds$, oddness gives $\int p d\mu = P$ and $\int n d\mu = 1 - Q$. For a function in $[0, 1]$ with prescribed mass, its s^2 moment is maximized by an upper threshold and minimized by a lower threshold: subtract the threshold and multiply by $s^2 - t^2$ to check the comparison. Under $s^2/2$ the measure is exponential with mean one, giving

$$\int s^2 p d\mu \leq 2P(1 - \log P), \quad \int s^2 n d\mu \geq 2(1 - Q + Q \log Q).$$

Subtracting proves $c \geq D(x, y)$. Retaining the exact linear cost Ly , the fiber estimate and identity cancellation give

$$\mathcal{B}_{\alpha_0, \kappa_0} - \kappa_0/\nu \leq \alpha_0 x^2 + Ly + R(2 - 3x + c) - c^2/6 + I + J.$$

Maximization over $c \geq D(x, y)$ yields

$$\alpha_0 x^2 + Ly + R(2 - 3x) + 3R^2/2 + I + J - [D(x, y) - 3R]_+^2/6.$$

For $x \geq 1/2$, this is increasing in y : one has $D \leq x \leq 1$ and $0 \leq \partial_y D = \log(Q/P) \leq \log(1/x) \leq \log 2 < 1$, so its derivative is at least $L - 1/3 \geq 0$. Substitute $y = 1 - x$, where $D = D_+(x)$. This substitution is for the full budget; it does not assert $c \geq D_+(x)$ for arbitrary sign pairs. $\square$

3.5 Exact fiber parameters

The following vectors specify the geometry of the indexed rows O_i . The index 15 is not used. The rows with $i \leq 14$ are clipped and those with $i \geq 16$ are ordinary. Additional rows and the complete family are specified in Section 3.5. All rows have cubic cost and $I = 0$. All vectors are indexed from one.

Shared settings. Set $\theta_i = 1$ for $i \geq 5$ and $\zeta_i = 1$ for $i \geq 7$. For the ordinary rows $16 \leq i \leq 47$, set

$$(\lambda_i, s_{o,i}, \epsilon_i, a_i) = (1, 0, 1, 0).$$

Thus only (r_i, L_i, R_i) vary in this family. Put $D = 10^{11}$ and set $\theta_i = \mathbb{T}_i/D$ for $i \leq 4$ and $\zeta_i = \mathbb{Z}_i/D$ for $i \leq 6$, where

$$\mathbb{T} = (18958859027, 36076041917, 44892200252, 57458657400).$$

$$\mathbb{Z} = (20244552040, 19381053236, 20244552040, 18804876939, 25318787960, 27111390000).$$

Clipped rows. For $1 \leq i \leq 15$, define

$$(\lambda_i, s_{o,i}, \epsilon_i, a_i) = (V_i, -W_i, E_i, A_i)/D,$$

using the four vectors

$$\begin{aligned} V &= (87761491006, 73599900674, 69508626533, 65504120567, 64758723783, 80738742719, \\ &\quad 93210555291, 98525115898, 99732601995, 99825884387, 99778401469, 99866625020, \\ &\quad 99954390397, 99979145755, 99985214444). \\ W &= (44492731254, 35343605792, 33914501943, 32848343951, 34962610096, 48613781674, \\ &\quad 74828246520, 122504286304, 182278769015, 182476544325, 155004678043, 166499657091, \\ &\quad 162280826361, 149988202073, 89995448017). \\ E &= (25174824009, 36397016861, 38651209454, 40582056575, 43477868822, 43991280471, \\ &\quad 43240623947, 33989061703, 17788209845, 37224392975, 211418201902, 397686648631, \\ &\quad 264254755411, 184701941749, 458969414100). \\ A &= (101392153639, 91982962402, 90351563064, 89046902177, 87180649505, 86607501450, \\ &\quad 86323045252, 91212824023, 104063934446, 82885592996, 31090792055, 19640755330, \\ &\quad 26054344215, 31627174773, 12257996126). \end{aligned}$$

The clipping reserve $J_i = (1 - \lambda_i)(1 + \epsilon_i)c_0 a_i^2$ is retained; only I vanishes. For the ordinary rows, $\gamma_i = J_i = 0$.

Matrix and cost parameters. For $1 \leq i \leq 47$, put $r_i = U_i/D$, where

$$\begin{aligned} U &= (6840508539, 7867500671, 8129698988, 8396663947, 11252275787, 19183692291, \\ &\quad 31158906782, 41195556046, 45688972577, 46621078147, 47228796419, 47526870166, \\ &\quad 47615490971, 47644291192, 47669575767, 47677676163, 47679103214, 47681135411, \\ &\quad 47683973044, 47687648033, 47692229857, 47697782305, 47701821764, 47704362221, \\ &\quad 47709157100, 47711998341, 47716819824, 47723374274, 47727906463, 47733269762, \\ &\quad 47739444266, 47746355658, 47751189228, 47753881155, 47759210512, 47761907563, \\ &\quad 47767581112, 47770196055, 47775221642, 47780103140, 47783063053, 47786259197, \\ &\quad 47789276700, 47792482837, 47794979488, 47797800404, 47800645876). \end{aligned}$$

Set $L_i = (\mathbf{L})_i - \Delta_i$ and $R_i = (\mathbf{R})_i$, with Δ_i specified below. The unscaled cost vectors are

$$\begin{aligned} \mathbf{L} &= (0.9501, 1.05841, 1.0779834, 1.0931, 1.19, 1.41, 1.61, 1.76, 1.836, 1.853, \\ &\quad 1.8629, 1.868, 1.869875, 1.870593, 1.871233, 1.871376, 1.871338, 1.871281, 1.871203, 1.871102, \\ &\quad 1.8709752, 1.870822, 1.87071, 1.87064, 1.87051, 1.87043, 1.87028871, 1.87011, 1.86998, 1.8698231, \\ &\quad 1.869648, 1.86945, 1.86931, 1.869232, 1.86908, 1.869, 1.86884, 1.86876, 1.86862, 1.86847, \\ &\quad 1.86839, 1.86829, 1.8682, 1.8681, 1.86802, 1.86794, 1.867858). \\ \mathbf{R} &= (0.2137, 0.1003, 0.0725852, 0.04715, 0.011, 0.03, 0.11, 0.2, 0.253, 0.265, 0.2731, 0.2773, \\ &\quad 0.27888, 0.2795, 0.28006, 0.2806, 0.28114, 0.28195, 0.2831, 0.28451, 0.28632, 0.28854, 0.2902, 0.2912, \\ &\quad 0.2932, 0.2943, 0.296217, 0.299, 0.301, 0.302993, 0.30557, 0.308471, 0.31053, 0.3117, 0.314, 0.3151, \\ &\quad 0.31754, 0.319, 0.32086, 0.323, 0.3243, 0.326, 0.3272, 0.3286, 0.32971, 0.331, 0.3322). \end{aligned}$$

The formulas determine all derived matrix and dual coefficients.

Cost corrections and additional rows. For the indexed rows use

$$\mathcal{I} = (1, 2, 3, 4, 5, 12, 13, 14, 16, 17, 18, 19, 20, 21, 47),$$

$$10^9(\Delta_i)_{i \in \mathcal{I}} = (93566, 3959, 264, 31127, 1087468, 16286, 3212, 2649, \\ 2631, 2523, 2332, 4474, 2423, 2017, 5769),$$

and $\Delta_i = 0$ otherwise. Denote row O_3 by O_3^+ ; thus its linear cost is 1.077983136. Add the following three clipped rows.

	A	B	C
θ	.42392200252	1	.47392200252
r	.08041574817	.47666345554	.08207765460
λ	.70401892810	.99990480883	.68658584120
s_o	-.34128247423	-1.41989395514	-.33700027130
ζ	.20244552040	1	.10244552040
ϵ	.38149440093	1.89778941067	.39102370079
a	.90701824607	.29993670991	.90035921632
L	1.07198986650	1.87106194702	1.08283940207
R	.07929603521	.27989192166	.06655654412

Add four further clipped rows, each with $\theta = \zeta = 1$.

	E	F	G	H
r	.47663384103	.47672651576	.47676121239	.47669394028
λ	.99989962564	.99978781552	.99990668028	.99980686411
s_o	-1.32690591529	-.39599881454	.20867774084	-.9638583209
ϵ	2.02874833711	4.66412751077	4.79251976568	4.64222990576
a	.27608922425	.08969262278	.05086123433	.1254500527
L	1.87108365601	1.87126980229	1.87137673056	1.87116437974
R	.27992051386	.28006288082	.28025534796	.28001573262

The final family consists of the following fifty-three rows, in this order:

$$O_1, O_2, A, O_3^+, C, O_4, \dots, O_{14}, B, E, H, F, G, O_{16}, \dots, O_{47}.$$

For every row reconstruct β, η, Q_e, Q_o , the dual coefficients, γ and J from the formulas above, always using $\alpha_0, \kappa_0, \rho, \sigma, c_0$. All rows have $I = 0$. The repair slopes will be generated by Lemma 3.11; they are not additional input data.

3.6 Exact full-domain certification

The unbounded exterior. Let $\mathcal{R}$ denote the left side of Eq. (13) minus its quadratic penalty, before I . Simultaneous sign changes $(A, B) \mapsto (-A, -B)$ and $(M, N) \mapsto (-M, -N)$ preserve it, so take $A, M \geq 0$. Put $S_0 = \sqrt{\pi}/2$, $K = 251/100$ and $\mu_0 = 1/4$. Dropping the nonnegative cost and using $\int |\phi| d\mu \leq 1$ gives

$$\mathcal{F}_{\kappa_0} \leq 2S_0(A + |B|) + 2(|M| + |N|), \quad 2S_0 < K.$$

Odd quadratic duality bounds $2(|M| + |N|) - (o_a M^2 + 2o_b MN + o_c N^2)$ by $g_a + g_c + 2|g_b|$. For clipped rows $H_o \leq (h_A A + h_B B)^2 / d_0$. The effective even penalty is

$$E_{\text{eff}} = \begin{pmatrix} e & p \\ p & q \end{pmatrix} - \frac{1}{d_0} \begin{pmatrix} h_A^2 & h_A h_B \\ h_A h_B & h_B^2 \end{pmatrix} = T_e^\top (Q_e + \gamma w_o w_o^\top)^{-1} T_e, \quad T_e = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The equality is the rank-one inverse-update identity. For ordinary rows omit the subtracted matrix and set $\gamma = 0$. Exact principal minors give $E_{\text{eff}} \succ \mu_0 I_2$ for every row. Hence

$$\mathcal{R} \leq K(A + |B|) - \mu_0(A^2 + B^2) + g_a + g_c + 2|g_b|.$$

Outside $[0, 16] \times [-16, 16]$ this is at most

$$16K - 256\mu_0 + K^2/(4\mu_0) + g_a + g_c + 2|g_b| < 0.$$

The last inequality is checked rationally for each row; the corresponding one-variable quadratic decreases beyond 16. It remains to cover

$$[0, 16] \times [-16, 16] \times [0, m_*] \times [-n_*, n_*]. \quad (16)$$

Near the origin, the positive-part integrals vanish on whole boxes; the next criterion detects these regions exactly.

Lemma 3.8. *On a four-variable box let $A_* = \max |A|$, $B_* = \max |B|$, $N_* = \max |N|$ and $O_* = \max |M - N|$, using coordinate endpoints and the four (M, N) corners. Put*

$$c_c = \kappa_0/\theta - A_*, \quad c_l = L/2 - O_*, \quad c_q = (R - N_*)/2.$$

If $c_c, c_l, c_q \geq 0$ and $B_^2 \leq 4c_l c_q$, and for a clipped row $|h_A A + h_B B| \leq a$ at all four even-coordinate corners, then $\mathcal{F}_{\kappa_0} = H_o = 0$ throughout the box and $\mathcal{R} \leq 0$.*

Proof. Every cubic in Eq. (12) is at most $-c_c - z(c_l - B_* z + c_q z^2)$ for $z \geq 0$. The quadratic is nonnegative by its discriminant; if $c_q = 0$, the hypothesis forces $B_* = 0$. Thus all positive parts vanish, including their unbounded tails. The affine hinge coordinate lies between its corner extrema, so the hinge also vanishes. The residual is the negative of positive-definite quadratic forms. $\square$

The criterion holds on a neighborhood of the origin, since every row has $\kappa_0, L, R > 0$ and every clipped row has $a > 0$. It handles the exact zero residual there without adding a positive reserve.

Continuous box bounds. Both $\mathcal{F}_{\kappa_0}$ and H_o are convex. Independent endpoint interpolation on a box gives

$$\mathcal{R} \leq \max_{16 \text{ vertices}} \mathcal{R} + (ed_A^2 + qd_B^2 + o_a d_M^2 + o_c d_N^2)/4. \quad (17)$$

The two cross terms interpolate exactly; a squared coordinate contributes its variance, at most a quarter of the squared side length. If all four even-coordinate corner values of $h_A A + h_B B$ are at least a , or all are at most $-a$, the hinge is one quadratic throughout the box. Cancel its quadratic part first, replacing e, q by $e - h_A^2/d_0, q - h_B^2/d_0$. These are positive by the effective penalty bound. All affine terms interpolate exactly. On a box crossing a hinge boundary use the unreduced coefficients in Eq. (17).

First accept any box satisfying Lemma 3.8. For every other box, accept only when its outward-rounded upper bound is strictly below $I = 0$. Otherwise bisect the coordinate with largest weighted squared side length, breaking ties in the order A, B, M, N , and process both halves, left first. Start the root at depth zero; failure at depth 110 is not an accepted box. Exhausting the subdivision stack proves complete coverage. Appendix A.2 specifies every interval operation and vertex enclosure, including multiple roots and Gaussian tails. For the cubic primitives one may use

$$e_z = e^{-z^2/2}, \quad E(z) = \int_0^z e^{-t^2/2} dt, \quad J_0 = E, \quad J_1 = -e_z, \quad J_2 = E - z e_z, \quad J_3 = -(z^2 + 2)e_z.$$

They satisfy $dJ_j/dz = z^j e^{-z^2/2}$. For higher-degree formulas in the general integration rules, $J_j = (j - 1)J_{j-2} - z^{j-1}e_z$.

3.7 Odd symmetrization and finite-matrix transfer

The odd part $f_o = (f - f \circ (-\text{id}))/2$ of any sign is ternary. Fill its zero set with either an odd measurable sign or its negative, leaving nonzero values unchanged. This expresses f_o as the average of two odd signs. Do likewise for g_o . All three terms of $\mathcal{B}_{\beta,\eta}$ are bilinear in the odd parts, so averaging the four odd-sign-pair inequalities removes the oddness assumption.

An antipodally paired partition transfers the Gaussian inequality to finite matrices while preserving the odd-projection term.

Lemma 3.9 (Transfer with an odd projection). *If $\mathcal{B}_{\alpha,\kappa}(f, g) \leq C + \kappa/\nu$ holds for all measurable signs in every finite positive dimension, where $\alpha, C, \kappa > 0$, then $K_G \geq (\pi\alpha/2 + \kappa/\nu)/(C + \kappa/\nu)$.*

Proof. Let $T_n = \alpha P_1 - P_3 + \nu\kappa P_{\text{odd}}$. Multiplication of the sign inequality by ν^2 gives $\langle f, T_n g \rangle \leq \nu^2 C + \nu\kappa$. For $v(x) = x/\|x\|$ away from zero, each coordinate is odd, so its vector objective is

$$V_n = \alpha \sum_i \|P_1 v_i\|_2^2 - \sum_i \|P_3 v_i\|_2^2 + \nu\kappa.$$

Rotational symmetry gives $\sum_i \|P_1 v_i\|_2^2 = (\mathbb{E}\|X\|)^2/n \rightarrow 1$. Also

$$\sum_i \|P_3 v_i\|_2^2 \leq \sum_i \|v_i - X_i/\sqrt{n}\|_2^2 = \mathbb{E}(1 - \|X\|/\sqrt{n})^2 \leq 2/n,$$

using $|\sqrt{t} - 1| \leq |t - 1|$ and $\text{Var}(\|X\|^2/n) = 2/n$. The same estimate implies the first convergence, so $V_n \rightarrow \alpha + \nu\kappa$.

To obtain finite matrices, approximate v uniformly, outside the null origin, by finitely many unit representatives u_i on an antipodally paired measurable partition $C_{-i} = -C_i$, $u_{-i} = -u_i$. Choose a finite spherical net on one measurable half of the antipodal pairs and extend by reflection. The finite-rank part has the integrable kernel

$$K_n(x, y) = \alpha \sum_{j=1}^n x_j y_j - \sum_{|\xi|=3} H_\xi(x) H_\xi(y),$$

where H_ξ are orthonormal multivariate Hermite. Define

$$A_{ij} = \int_{C_i \times C_j} K_n(x, y) d\gamma_n(x) d\gamma_n(y) + \frac{\nu\kappa}{2} \gamma_n(C_i) (\mathbf{1}_{j=i} - \mathbf{1}_{j=-i}).$$

Each cell signing gives measurable f, g . The second matrix term is $\nu\kappa \langle P_{\text{odd}} f, P_{\text{odd}} g \rangle$, so its scalar objective is bounded by $\nu^2 C + \nu\kappa$. For the antipodal vector representatives its contribution is exactly $\nu\kappa$. The finite Grothendieck inequality therefore bounds the partition vector objective by $K_G(\nu^2 C + \nu\kappa)$. Integrability and uniform approximation pass this to V_n . Letting $n \rightarrow \infty$ gives $\alpha + \nu\kappa \leq K_G(\nu^2 C + \nu\kappa)$, which is the stated ratio. This construction does not treat P_{odd} as an integrable kernel. $\square$

3.8 Retained remainders and a shared Bessel budget

We retain the two matrix remainders which act on the first-level disagreement coordinates. All matrices use the reference parameters α_0, κ_0 of Section 3.

A weighted Bessel estimate links the two disagreement components to their shared cubic moment.

Lemma 3.10 (Weighted Bessel inequality). *With the raw conditional moments $K_j(Y)$ and transverse projections Π_ℓ used above, put*

$$u = (u_0, u_2) = \frac{1}{\nu}(\Pi_1 K_0, \Pi_1 K_2), \quad v = (v_1, v_3) = \frac{1}{\nu}(\mathbb{E}K_1, \mathbb{E}K_3).$$

Then $q_1 = \|u_0\|_2^2 + v_1^2$ and

$$\|u_0 + u_2\|_2^2 + \frac{1}{2}(3v_1 + v_3)^2 \leq z. \quad (18)$$

Proof. The identity for q_1 follows from raw-moment orthogonality. If $Y = (Y_1, \dots, Y_d)$, the functions

$$\phi_j = |S|^{1/2} Y_j / \sqrt{\nu} \quad (1 \leq j \leq d), \quad \phi_0 = \text{sgn}(S) |S|^{3/2} / \sqrt{2\nu}$$

are orthonormal in the full Gaussian space, since $\mathbb{E}|S| = \nu$, $\mathbb{E}|S|^3 = 2\nu$ and the transverse coordinates are independent centered standard Gaussians. Put $W = |S|^{3/2} k$. The identities $1 + \text{He}_2(S) = S^2$ and $3 \text{He}_1(S) + \text{He}_3(S) = S^3$ give

$$\|u_0 + u_2\|_2^2 = \nu^{-1} \sum_{j=1}^d |\langle W, \phi_j \rangle|^2, \quad (3v_1 + v_3)^2 / 2 = \nu^{-1} |\langle W, \phi_0 \rangle|^2.$$

Bessel's inequality bounds their sum by $\|W\|_2^2 / \nu = \mathbb{E}[|S|^3 k^2] / \nu = z$. In zero transverse dimension the first sum is empty, so the same argument applies. $\square$

The matrix remainders absorb a small first-level perturbation at a cost controlled by the shared moment.

Lemma 3.11 (Generated matrix repairs). *For each row, define the rank-one positive-semidefinite remainders*

$$E_i = \theta_i(Q_{e,i} - A_{e,i}), \quad O_i^{\text{mat}} = \theta_i(Q_{o,i} - \text{diag}(-\beta_i, 1/6)).$$

Write their entries as $(a_e, b_e; c_e)$ and $(a_o, b_o; c_o)$, and put

$$e_0 = 1/1000, \quad d_e = a_e - 2b_e + c_e - e_0, \quad d_o = a_o - 6b_o + 9c_o - e_0.$$

The printed rows satisfy $c_e, c_o, d_e, d_o > 0$. Define

$$\chi_i = \frac{1 + \lfloor 1000 \max\{c_e/d_e, 2c_o/d_o\} \rfloor}{1000}.$$

For $0 \leq e \leq e_0$,

$$eq_1 - \langle u, E_i u \rangle - v^\top O_i^{\text{mat}} v \leq \chi_i e z. \quad (19)$$

Here and below the first quadratic form is integrated over Y .

Proof. The matrix comparisons give determinant zero and positive bottom-right entries. Exact substitution of the printed rational parameters gives $d_e > 1.93$ and $d_o > 1.64$ for every row; these are the only extra sign checks needed for the formula. For $0 < e \leq e_0$, direct expansion gives

$$\det(E_i + \chi_i e(1, 1)^\top (1, 1) - e \text{diag}(1, 0)) = e\{\chi_i(a_e - 2b_e + c_e - e) - c_e\} > 0,$$

$$\det(O_i^{\text{mat}} + \chi_i e(3, 1)^\top (3, 1)/2 - e \text{diag}(1, 0)) = e\{\chi_i(a_o - 6b_o + 9c_o - e)/2 - c_o\} > 0.$$

The equalities use the zero determinants of the original remainders. The strict inequalities follow from $\chi_i > \max\{c_e/d_e, 2c_o/d_o\}$ and $e \leq e_0$. Both bottom-right entries are positive, so both repaired matrices are positive definite. At $e = 0$ use the original semidefinite comparisons. Apply the first matrix to u and integrate, and the second to v . Eq. (18) bounds their combined repair cost by $\chi_i e z$, proving Eq. (19). $\square$

Keeping these remainders in the fiber and phase accounting preserves the terms needed for the parameter refinement.

Lemma 3.12 (Retained row inequality). *Each certified row satisfies*

$$\begin{aligned} \mathcal{A}_{\alpha_0, \kappa_0} \leq & [\alpha_0 - (1 - \theta_i)\rho]x^2 + (1 - \theta_i)a_0 + \theta_i L_i y + \theta_i R_i z + \theta_i J_i \\ & - (2 - \theta_i)p_3 - \langle u, E_i u \rangle - v^\top O_i^{\text{mat}} v. \end{aligned} \quad (20)$$

Proof. In Lemma 3.3, retain the first odd-comparison remainder at transverse degree zero and the first even-comparison remainder at transverse degree one. After division by ν^2 , these are the corresponding quadratic forms in u, v . Mix the fiber and phase inequalities with weights $\theta_i, 1 - \theta_i$, as in Lemma 3.6. The identities

$$\theta_i \beta_i - \alpha_0 - (1 - \theta_i)\rho = 0, \quad \theta_i \eta_i - (1 - \theta_i)\sigma = 0$$

cancel q_1, z_5 . Since the signs are odd, $\langle f, g \rangle = 1 - 2m$, so the constant fiber cost $2\kappa_0 m/\nu$ cancels the shifted odd-projection term. Discard only the nonpositive agreement fifth-degree term and retain p_3, y, z . This gives Eq. (20). $\square$

3.9 Shared disagreement-moment bounds

A conical rearrangement sharpens the even part of the shared Bessel budget.

Lemma 3.13 (Sharp conical budget). *Let S be standard Gaussian and Y an independent standard Gaussian vector. For every measurable ternary $k \in \{-1, 0, 1\}$,*

$$\frac{\|\mathbb{E}[S^2 Y k]\|^2}{\nu^2} \leq (12 - 8\sqrt{2}) \frac{\mathbb{E}[|S|^3 k^2]}{\nu}.$$

The coefficient is sharp in every positive transverse dimension.

Proof. If $\mathbb{E}[S^2 Y k] = 0$ or the weighted mass is zero, the assertion is immediate; this includes zero transverse dimension. Otherwise project Y onto the direction of $\mathbb{E}[S^2 Y k]$, obtaining a standard Gaussian Z independent of S . The ternary identity $|k| = k^2$ gives

$$\|\mathbb{E}[S^2 Y k]\| \leq \mathbb{E}[S^2 |Z| k^2].$$

Condition on (S, Z) and put $p = \mathbb{E}[k^2 | S, Z] \in [0, 1]$. Under the finite measure $|S|^3 d\gamma_2$, fix the mass $B = \mathbb{E}[|S|^3 p]$. A threshold $a \geq 0$ with the same mass in $\{|Z| > a|S|\}$ exists by continuity; the zero mass is obtained as a limit. Away from a null set,

$$(|Z|/|S| - a)(p - 1_{\{|Z| > a|S|\}}) \leq 0.$$

Integrating cancels the equal masses. Thus the conical indicator maximizes $\mathbb{E}[S^2 |Z| p]$ at fixed B .

Write $a = \tan \theta$, $s = \sin \theta$. Polar integration gives

$$A_s = \nu(1 - s^2)^{3/2}, \quad B_s = \nu(2 - 3s + s^3),$$

for the objective and weighted mass of the cone. Indeed the radial moment is $\mathbb{E}[R^3] = 3/\nu$ and the angular integrals are $(2/\pi) \int_{\theta}^{\pi/2} \cos^2 u \sin u du$ and $(2/\pi) \int_{\theta}^{\pi/2} \cos^3 u du$. Therefore

$$\frac{A_s^2}{\nu B_s} = F(s) := \frac{(1 - s)(1 + s)^3}{s + 2}, \quad \frac{dF}{ds} = -\frac{3(s + 1)^2(s^2 + 2s - 1)}{(s + 2)^2}.$$

Its maximum on $[0, 1]$ is $F(\sqrt{2} - 1) = 12 - 8\sqrt{2}$. Equality holds for $k = \text{sgn}(Z)1_{\{|Z| > a|S|\}}$ at this threshold, proving sharpness. $\square$

The extremizer is compatible with odd agreement variables: take $h = \text{sgn}(S)$ on the complementary cone and zero on the cone, and put $f = h + k$, $g = h - k$. Then f, g are odd signs and S is the direction of $P_1 h$.

The same rearrangement gives stronger moment bounds when the weighted disagreement mass is small.

Lemma 3.14 (Nonlinear moment budgets). *For the same ternary function put*

$$z = \mathbb{E}[|S|^3 k^2]/\nu, \quad z_e = \|\mathbb{E}[S^2 Y k]\|^2/\nu^2, \quad z_o = (\mathbb{E}[S^3 k])^2/(2\nu^2).$$

Then $0 \leq z \leq 2$ and

$$z_e \leq \frac{8}{3\sqrt{3}} z^{3/2} \leq 12z^2 + \frac{4}{81} z, \quad z_o \leq z^2/2.$$

Proof. The conical rearrangement in Lemma 3.13 maximizes the even objective at each fixed weighted mass. Its formulas give, for $0 \leq s < 1$,

$$z = (1 - s)^2(s + 2), \quad z_e \leq (1 - s^2)^3,$$

and therefore

$$\frac{z_e}{z^{3/2}} \leq \frac{(1 + s)^3}{(s + 2)^{3/2}} \leq \frac{8}{3\sqrt{3}}.$$

The second inequality follows by raising the positive quantities to the power $2/3$ and using $3(s + 1)^2 - 4(s + 2) = (s - 1)(3s + 5) \leq 0$. The zero-mass case follows directly, and the full-mass case is $s = 0$. The argument includes zero transverse dimension, when $z_e = 0$. The arithmetic-geometric mean inequality gives $12z + 4/81 \geq (8/(3\sqrt{3}))\sqrt{z}$, since $4 \cdot 12 \cdot (4/81) = 64/27$; multiplication by z proves the rational majorant. Finally $|k| = k^2$ gives $|\mathbb{E}[S^3 k]| \leq \mathbb{E}[|S|^3 k^2] = \nu z$, proving the odd bound. $\square$

We next bound disagreement mass using its first and third absolute moments.

Lemma 3.15 (Two-moment mass dual). *For $0 < a < b$, define*

$$\lambda = \frac{a^2 + ab + b^2}{ab(a + b)}, \quad \kappa_d = \frac{1}{ab(a + b)}, \quad P(s) = 1 - \lambda s + \kappa_d s^3.$$

For every measurable $0 \leq v \leq 1$, writing $m_v = \mathbb{E}[v]$, $y_v = \mathbb{E}[|S|v]/\nu$, $z_v = \mathbb{E}[|S|^3 v]/\nu$, one has

$$m_v/\nu \leq \lambda y_v - \kappa_d z_v + Z_{a,b}, \quad Z_{a,b} = \int_0^a P(s) e^{-s^2/2} ds + \int_b^\infty P(s) e^{-s^2/2} ds.$$

Proof. Condition on $|S|$ to obtain $p(s) \in [0, 1]$. Pointwise,

$$p(s) \leq \lambda s p(s) - \kappa_d s^3 p(s) + [P(s)]_+.$$

The exact factorization

$$P(s) = (1 - s/a)(1 - s/b)(1 + s/(a + b))$$

shows that its nonnegative set on the nonnegative axis is $[0, a] \cup [b, \infty)$. Integrate against $e^{-s^2/2} ds$ and use the density $\nu e^{-s^2/2}$ of $|S|$. This proves the assertion on the full space, including endpoint masses zero and one. $\square$

We use only the following two fixed pairs, not a numerical solution of a varying moment problem:

$$(a_1, b_1) = (1.3124, 2.1229), \quad (a_2, b_2) = (.3349, 4.5446).$$

They have rational parameters

$$(\lambda_1, \kappa_{d,1}) = \frac{1}{2392767145197}(2253798032500, 2500000000000),$$

$$(\lambda_2, \kappa_{d,2}) = \frac{1}{742653332193}(2228753371000, 1000000000000).$$

The exact upper caps are

$$\overline{Z}_1 = .5315084750805476152266157064634276146723,$$

$$\overline{Z}_2 = .1654951631501003383927538130093105336710.$$

Here is a finite rational verification. Write $E(t) = \int_0^t e^{-s^2/2} ds$, $e_t = e^{-t^2/2}$, $S_0 = \sqrt{\pi/2}$. Then

$$Z_{a,b} = E(a) + S_0 - E(b) - \lambda(1 - e_a + e_b) + \kappa_d\{2 - (a^2 + 2)e_a + (b^2 + 2)e_b\}.$$

Use the odd/even exponential Taylor sums of degrees 199, 200 at $t^2/2$ and their integrated versions for $E(t)$. They enclose the functions even when initial terms increase. Use the Machin bounds of Appendix A.2 and outward integer square comparisons on the grid $10^{-70}\mathbb{Z}$ to enclose S_0 . Substitution proves $Z_{a_j, b_j} < \overline{Z}_j$. Thus no root isolation or numerical quadrature is needed for these two new duals.

A polynomial hierarchy supplies a complementary mass bound using only the first absolute moment.

Lemma 3.16 (A finite mass hierarchy). *Set $c_1 = 1/3$ and, for $2 \leq k \leq 8$, define*

$$c_k = \frac{2}{2k+1} \sum_{j=1}^{k-1} \frac{j c_j}{(k-j)(k-j+1)}, \quad H_8(y) = 2y - 2 \sum_{j=1}^8 \frac{c_j}{j+1} y^{j+1}.$$

For $0 \leq v \leq 1$, $(\mathbb{E}[v]/\nu)^2 \leq H_8(\mathbb{E}[|S|v]/\nu)$. Moreover H_8 is increasing on $[0, 1]$.

Proof. Let $p(s) = \mathbb{E}[v \mid |S| = s]$. For a prescribed mass $\mathbb{E}[v]/\nu = I(t) = \int_0^t e^{-s^2/2} ds$, moving mass to the origin minimizes its weighted first moment. Explicitly integrate $(s-t)(p(s) - 1_{[0,t]}(s)) \geq 0$. Thus $y_v \geq V(t) := 1 - e^{-t^2/2}$. Put $f(y) = 1 - \sum_{j=1}^8 c_j y^j$, $J(t) = t f(V(t)) - I(t)$. Since

$$(1-y) \log(1-y) = -y + \sum_{k \geq 2} \frac{y^k}{k(k-1)},$$

we have

$$\frac{dJ}{dt} = y - \sum_j c_j y^j + 2(1-y) \log(1-y) \sum_j j c_j y^{j-1}, \quad y = V(t).$$

The recurrence cancels its coefficients of degrees one through eight; every remaining coefficient is nonnegative. Since $J(0) = 0$, this gives $J \geq 0$. If $F(t) = H_8(V(t)) - I(t)^2$, then $dF/dt = 2e^{-t^2/2} J(t) \geq 0$ and $F(0) = 0$. All c_j are positive and the exact sum is $446906/883575 < 1$, so $dH_8/dy > 0$ on $[0, 1]$. This proves the assertion, with masses zero and one obtained by limits. $\square$

The three scalar bounds. Retain Eq. (20) and the shared Bessel budget. We sharpen only the analytic phase and clipping constants:

$$a_+ = 1.24607768922, \quad \rho_+ = .9418458229699, \quad c_+ = .793277694864.$$

Here is an exact check from Lemma 3.1. Put $u = \cos(\pi/7)$ and $v = \sin(\pi/14)$. The Machin enclosure of π and the odd/even sine and cosine Taylor sums through indices 9, 10 give

$$\frac{2\pi}{7(u^3 - v^3)} < a_+, \quad \rho < \rho_+ < \frac{1}{u^2 + uv + v^2},$$

$$\frac{4}{7(u^3 - v^3)} < c_+ < c_0, \quad \sigma < \frac{u^5 - v^5}{u^3 - v^3}.$$

All arguments are below one, so alternating remainders enclose the trigonometric functions. The same seven-copy phase proof applies with a_+, ρ_+ , and the old clipping operator still has squared norm at most $c_+ a^2$ on odd $|F| \leq a$, since its first-level weight is the smaller ρ .

Set $J_i^+ = (1 - \lambda_i)(1 + \epsilon_i)c_+ a_i^2$. This changes only the analytic clipping reserve, not the hinge penalty or any fiber cost. The matrix identities still use the original $\rho, \sigma, \alpha_0, \kappa_0$. Repeating the row accounting leaves an additional nonpositive term $-(1 - \theta_i)(\rho_+ - \rho)q_1$, which may be discarded. Thus

$$\mathcal{A}_{\alpha_0, \kappa_0} \leq [\alpha_0 - (1 - \theta_i)\rho_+]x^2 + (1 - \theta_i)a_+ + \theta_i L_i y + \theta_i R_i z + \theta_i J_i^+ \\ - (2 - \theta_i)p_3 - \langle u, E_i u \rangle - v^\top O_i^{\text{mat}} v.$$

The fifth-level and constant fiber costs cancel as before. No fiber matrix is reconstructed at ρ_+ .

Mixing the retained matrices before repair allows their determinants to reduce the perturbation cost.

Lemma 3.17 (Mixed moment repairs). *For convex row weights put $E = \sum_i w_i E_i$, $O = \sum_i w_i O_i^{\text{mat}}$, with entries $(a_e, b_e; c_e)$ and $(a_o, b_o; c_o)$. For $e > 0$ suppose c_e, c_o and the denominators below are positive. Define*

$$g_e = \frac{[ec_e - \det E]_+}{a_e - 2b_e + c_e - e}, \quad g_o = \frac{2[ec_o - \det O]_+}{a_o - 6b_o + 9c_o - e},$$

$$c_* := 6863/10000.$$

Then $eq_1 - \langle u, Eu \rangle - v^\top Ov \leq \Gamma z + Dz^2$ for either of the two choices

$$(\Gamma, D) = (g_o + c_*[g_e - g_o]_+, 0), \quad (\Gamma, D) = (4g_e/81, 12g_e + g_o/2).$$

We call these the conical and power repairs, respectively.

Proof. The rational test $(12 - c_*)^2 < 128$, together with $12 - c_* > 0$, proves $c_* > 12 - 8\sqrt{2}$. Set $z_e = \|u_0 + u_2\|_2^2$, $z_o = (3v_1 + v_3)^2/2$. Lemma 3.10 gives $z_e + z_o \leq z$, and Lemma 3.13 gives $z_e \leq c_* z$, since $u_0 + u_2$ represents $\mathbb{E}[S^2 Y k]/\nu$.

Direct determinants and positive bottom-right entries give

$$E + g_e \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - e \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0, \quad O + \frac{g_o}{2} \begin{pmatrix} 9 & 3 \\ 3 & 1 \end{pmatrix} - e \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0.$$

Their determinants are $\det E - ec_e + g_e(a_e - 2b_e + c_e - e)$ and $\det O - ec_o + g_o(a_o - 6b_o + 9c_o - e)/2$, respectively. Apply the first matrix to u and integrate, and apply the second to v . The resulting bound is $g_e z_e + g_o z_o$. If $g_e \geq g_o$, the shared and conical budgets give $[g_o + c_*(g_e - g_o)]z$; otherwise the shared budget gives $g_o z$. This proves the conical repair. For the power repair apply Lemma 3.14 to both terms at the same actual weighted mass z . $\square$

Mixing before repair retains the possibly positive determinants of E, O . The affine repair $(\Gamma, D) = (e \sum_i w_i \chi_i, 0)$ remains valid and will be used at the right endpoint.

Use

$$\begin{aligned}\epsilon_+ &= .000001715757, & \delta_+ &= .000000294200, & \zeta_+ &= .00000000862054, \\ \alpha_+ &= \alpha_0 - \epsilon_+ = 1.433927284243, & \kappa_+ &= \kappa_0 - \delta_+ = .0079949058, \\ C_+ &= \alpha_+ - 1/6 + \zeta_+ = \frac{190089093929531}{150000000000000}.\end{aligned}$$

At $e = \epsilon_+$ the two repair denominators and c_e, c_o are positive for every row, by exact rational substitution. They are affine in the row weights, so they remain positive for every convex mixture. Subtracting $\epsilon_+ b_1 = \epsilon_+(x^2 - q_1)$ and applying a repair adds $\Gamma z + Dz^2 - \epsilon_+ x^2$. The later change in κ_+ adds exactly $2\delta_+ m/\nu$.

The repair and mass estimates now combine into one scalar majorant for each convex mixture of rows.

Lemma 3.18 (A common scalar majorant). *For convex row weights and any of the valid repairs (Γ, D) above, put*

$$(t, \ell, b_0, j) = \sum_i w_i(\theta_i, \theta_i L_i, \theta_i R_i, \theta_i J_i^+).$$

For $s = 1, 2$ use the fixed mass-dual parameters above. For $s = 3$ and $a > 0$ put

$$\lambda_3 = \kappa_{d,3} = 0, \quad Z_3(x) = \frac{H_8(1-x) + a^2}{2a}, \quad Z_s(x) = \bar{Z}_s \quad (s = 1, 2).$$

Define

$$L_s = \ell + 2\delta_+ \lambda_s, \quad R_s = b_0 + \Gamma - 2\delta_+ \kappa_{d,s}, \quad J_s(x) = j + 2\delta_+ Z_s(x).$$

Put $a_t = (2-t)/6$ and $z_0 = 2 - 3x$. If $L_s, R_s, D \geq 0$ and $a_t > D$, then $\mathcal{A}_{\alpha_+, \kappa_+} \leq U_s(x)$, where

$$\begin{aligned}U_s(x) &= [\alpha_+ - (1-t)\rho_+]x^2 + (1-t)a_+ + L_s(1-x) + R_s z_0 + Dz_0^2 + J_s(x) \\ &\quad + \frac{(R_s + 2Dz_0)^2}{4(a_t - D)}.\end{aligned}$$

In mode three write $U_{3,a} = U_3$.

Proof. Mix the retained row inequalities, subtract $\epsilon_+(x^2 - q_1)$ and apply the repair. The change from κ_0 to κ_+ adds exactly $2\delta_+ m/\nu$. For $s = 1, 2$, the two-moment dual bounds this by $2\delta_+(\lambda_s y - \kappa_{d,s} z + \bar{Z}_s)$. For $s = 3$, Lemma 3.16 and $y \leq 1 - x$ give $m/\nu \leq \sqrt{H_8(1-x)} \leq Z_3(x)$; the last inequality follows from $(\sqrt{H_8(1-x)} - a)^2 \geq 0$. Thus the remaining moment terms are $L_s y + R_s z + Dz^2 + J_s(x) - (2-t)p_3$. Use $y \leq 1 - x$, $z \leq z_0 + c$ and $p_3 \geq c^2/6$. The substitution in $R_s z + Dz^2$ is valid because $z_0 + c \geq z \geq 0$ and $R_s, D \geq 0$. Complete the square in the single shared variable c :

$$(R_s + 2Dz_0)c - (a_t - D)c^2 \leq \frac{(R_s + 2Dz_0)^2}{4(a_t - D)}.$$

The positive denominator justifies this maximization over all real c . In particular, the negative cubic term of the mass dual is incorporated before the common z bound, not charged to a second independent budget. $\square$

A finite cover selects mixtures and moment bounds that control the interior range of x .

Lemma 3.19 (Deterministically generated scalar cover). *The preceding bounds prove $\mathcal{A}_{\alpha_+, \kappa_+} < C_+$ for every $x \in [1/5, 999/1000]$.*

Exact rational algorithm and execution. Use the fifty-three rows in their defining order. For each adjacent pair, assign weight $1 - k/16384$ to the earlier row and $k/16384$ to the later row, where $k \in \{0, \dots, 16384\}$; all other weights are zero. Start with $[l, r] = [1/5, 999/1000]$. Set

$$a = 2^{-48}(1 + \lfloor 2^{48} \sqrt{H_8(1 - (l + r)/2)} \rfloor).$$

The floor is obtained by integer square comparisons. Test modes 1, 2, 3 in that order, adjacent row pairs in their printed order, and then the conical and power repairs in that order, at $e = \epsilon_+$. For a fixed pair, mode and repair, let $F(k)$ be the maximum of its two endpoint values. Nominate a weight by exact adjacent comparisons: start $L = 0, H = 16384$; at $k = \lfloor (L + H)/2 \rfloor$, replace H by k if $F(k) \leq F(k + 1)$, otherwise replace L by $k + 1$. Continue until $L = H$. This deterministic rule only nominates a candidate; neither convexity in the weight nor optimality of the nominee is a premise.

Accept only if the mixed matrix comparisons hold, $L_s, R_s, D \geq 0$, $a_t > D$, both endpoint values are strictly below C_+ , and the bound is convex in x . For modes one and two its second derivative is

$$2[\alpha_+ - (1 - t)\rho_+] + 18D + \frac{18D^2}{a_t - D} > 0.$$

In mode three use the uniform lower bound

$$2[\alpha_+ - (1 - t)\rho_+] + 18D + \frac{18D^2}{a_t - D} - \frac{2\delta_+}{a} \sum_{j=1}^8 jc_j > 0.$$

Otherwise continue the pair/mode search; if no choice passes, bisect the interval and process both children left first. An unresolved interval at depth 60 is failure.

This exact procedure is executed without reading any supplied witnesses. It exhausts its stack with 97 consecutive intervals, 194 accepted endpoint comparisons and maximum depth 20. Mode counts are 59, 34, 4; 90 intervals use conical repair and 7 use power repair. Every accepted endpoint gap exceeds $9.1854 \cdot 10^{-14}$, and every lower bound on the second derivative exceeds 1.34128936. A separate polynomial assembly rechecks every accepted polynomial and endpoint. Convexity bounds the whole interval by its larger endpoint value. The accepted intervals partition the starting interval, proving the claim. $\square$

The scalar cover handles the interior, while separate phase and support estimates control the two endpoint ranges.

Theorem 3.20 (Lower bound, formal version of Theorem 1.2). *For every finite positive Gaussian dimension and all measurable signs,*

$$\alpha_+ b_1 - b_3 + \frac{\kappa_+}{\nu} \langle P_{\text{odd}} f, P_{\text{odd}} g \rangle \leq C_+ + \kappa_+ / \nu. \quad (21)$$

Consequently,

$$K_G \geq \frac{107544546318225\pi + 1199235870000\sqrt{\pi/2}}{190089093929531 + 1199235870000\sqrt{\pi/2}} = 1.7712846852087007 \dots \quad (22)$$

Proof. Lemma 3.19 handles $1/5 \leq x \leq 999/1000$. For $0 \leq x \leq 1/5$, the direct phase bound gives

$$\mathcal{A}_{\alpha_+, \kappa_+} \leq a_+ + (\alpha_+ - \rho_+)/25 < C_+,$$

which is an exact rational comparison. For $999/1000 \leq x \leq 1$, use row O_{47} and put

$$d = 1 - x, \quad L_0 = 2\alpha_0 - 1, \quad \mu = .000005769, \quad L = L_0 - \mu, \quad R_0 = .3322.$$

The affine repair remains valid, so use $\Gamma = .06\epsilon_+$ at this endpoint. The support comparison uses $D_+(x) = x + 2x \log x$ and

$$D_4(x) = 1 - 3d + d^2 + d^3/3 + d^4/6 \leq D_+(x).$$

For clarity the underlying comparison $c \geq D(x, y)$ follows by rearranging the positive and negative parts of h under the probability measure $se^{-s^2/2}ds$. With $P = (1 - y + x)/2$, $Q = (1 + y + x)/2$, upper and lower exponential thresholds give

$$c \geq D(x, y) = x + 2(P \log P + Q \log Q).$$

By Lemma 3.7, after maximizing over $c \geq D(x, y)$, the entire y -dependent expression is increasing: $D \leq 1$, $0 \leq \partial_y D \leq \log 2 < 1$, and $L > 1/3$. Therefore substituting $y = 1 - x$ in this *whole budget* is valid. It does not assert $c \geq D_+(x)$ for arbitrary sign pairs.

Define the algebraic comparison

$$U_0(x) = \alpha_0 x^2 + L_0(1 - x) + R_0(2 - 3x) + 3R_0^2/2 - [D_+(x) - 3R_0]_+^2/6.$$

No separate fiber inequality at cost L_0 is asserted. With $w = 1 - 3R_0$, exact expansion gives $C_0 - U_0(x) \geq d^2 F(d)$, where

$$F(d) = 11/6 - \alpha_0 - R_0 + (w/9 - 1)d + (w/18 - 1/6)d^2 - d^3/18 + 2d^4/27 + d^5/54 + d^6/216.$$

The polynomial D_4 decreases on $0 \leq d \leq 1/1000$. The exact test $D_4|_{d=1/1000} > 3(R_0 + \Gamma)$ therefore activates both endpoint corrections. Expanding the squares gives

$$U_{\epsilon_+, \mu}(x) = U_0(x) - \epsilon_+ x^2 - \mu(1 - x) + \Gamma(2 - 3x + D_+(x)).$$

The positive series gives $2 - 3x + D_+(x) \leq d^2/(1 - d) \leq 1000d^2/999$. The elementary mass bound $m/\nu \leq \sqrt{2d}$ follows from Lemma 3.16. Put $v = \sqrt{2d}$ and $f_0 = 11/6 - \alpha_0 - R_0$. Retain the displayed terms of F through degree three and discard only its positive higher terms. Substitution gives

$$\mathcal{A}_{\alpha_+, \kappa_+} - (\alpha_+ - 1/6) \leq P(v),$$

where the rational polynomial is

$$\begin{aligned} P(v) = & 2\delta_+ v + (\epsilon_+ - \mu/2)v^2 - \frac{f_0 + \epsilon_+ - 1000\Gamma/999}{4}v^4 \\ & + \frac{1 - w/9}{8}v^6 + \frac{1/6 - w/18}{16}v^8 + \frac{1}{576}v^{10}. \end{aligned}$$

The actual interval has $0 \leq v = \sqrt{2d} < 1/20$. We prove $P < \zeta_+$ on the larger interval $[0, 1/20]$ by concavity. In exact rational form,

$$P(v) = a_1 v + a_2 v^2 - A v^4 + B v^6 + C v^8 + D v^{10},$$

$$a_1 = .000000588400, \quad a_2 = -.000001168743, \quad A = \frac{2486620003549}{148000000000000},$$

$$B = \frac{44983}{360000}, \quad C = \frac{14983}{1440000}, \quad D = \frac{1}{576}.$$

These satisfy $A > 1/60$, $0 < B < 1/8$ and $0 < C < 1/96$. Since $v^2 \leq 1/400$ and

$$\frac{30}{3200} + \frac{56}{15360000} + \frac{90}{36864000000} < \frac{1}{100},$$

we have $30Bv^4 + 56Cv^6 + 90Dv^8 \leq v^2/100$. Consequently,

$$\frac{d^2P}{dv^2} \leq -.000002337486 - (12A - 1/100)v^2 < -2 \cdot 10^{-6}.$$

The first inequality bounds the three positive derivative terms; the second uses $A > 1/60$. Take $v_0 = 20077419/10^9$ and $t = v_0^2 = 403102753701561/10^{18}$. Two rational substitutions give

$$P(v_0) = a_1v_0 + t\{a_2 + t[-A + t(B + t(C + tD))]\} < 8.620510323 \cdot 10^{-9},$$

$$\left| \frac{dP}{dv}(v_0) \right| = |a_1 + v_0\{2a_2 + t[-4A + t(6B + t(8C + 10Dt))]\}| < 7.72 \cdot 10^{-14}.$$

Both comparisons follow by clearing positive denominators. Taylor's theorem and completing the square now give

$$P(v) \leq P(v_0) + \frac{((dP/dv)(v_0))^2}{4 \cdot 10^{-6}} < 8.620510324 \cdot 10^{-9} < \zeta_+.$$

The first step uses the second-derivative bound, the second uses the two rational comparisons, and the final gap is $7419/(25 \cdot 10^{16}) > 0$. This includes $v = 0$ and proves the endpoint bound without a subdivision certificate.

Independent odd symmetrization of the two signs gives Eq. (21), by the odd-symmetrization argument in Section 3. All three parameters are positive, so Lemma 3.9 gives

$$K_G \geq \frac{\pi\alpha_+/2 + \kappa_+/\nu}{C_+ + \kappa_+/\nu}.$$

Substitution gives Eq. (22). The normalizer enclosures in Appendix A.2 certify its displayed decimal. $\square$

The conical and nonlinear budgets strengthen the shared Bessel estimate and permit mixing the retained matrices before repairing them. The mass duals, hierarchy and nonlinear moment bounds are analytic, while the fiber integrations and finite scalar cover are computer-assisted. The reserve ζ_+ remains positive; the final intercept is not asserted to be halfspace-sharp.

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A Mathematical specification of the finite verification

This appendix specifies the computer-assisted parts as mathematical finite checks. It is independent of a programming language or an interval library. The displayed data state inequalities, not floating-point equalities. An interval check succeeds only if its entire enclosure satisfies the stated inequality. The zero-region criterion of Lemma 3.8 is checked by exact rational comparisons. An unresolved interval, exhausted subdivision budget, or zero-containing divisor is a failure. No sampling argument is used.

A.1 Rational interval operations

For an integer precision p let $u_p = 2^{-p}$ and set

$$\text{out}_p([a, b]) = [u_p \lfloor a/u_p \rfloor, u_p \lceil b/u_p \rceil].$$

Rational inputs are enclosed by this operation. Addition and subtraction use their endpoint formulas. For multiplication take the minimum and maximum of the four endpoint products and apply out_p ; for division do the same with the four quotients, after checking that the divisor does not contain zero. For a nonnegative interval take square roots of both endpoints and round outwards; the bounds are determined by integer square comparisons. Nonnegative integer powers may be evaluated by repeated squaring. Complex intervals are rectangles, with the usual real formulas for arithmetic, conjugation and modulus. All intervening operations are enclosed, including the formation of derived inputs. Arbitrarily precise ball arithmetic is an equivalent realization if it proves the same endpoint inequalities.

A second realization uses directed binary floating-point endpoints with p significand bits: each nonzero endpoint is of the form $\pm m2^e$, where m is an integer with at most p binary digits. Each endpoint operation is rounded to the nearest representable number in the indicated direction: down for a lower bound and up for an upper bound. Interval division is realized by an outward-rounded reciprocal followed by interval multiplication. This realizes the same enclosing inequalities as the fixed absolute grid, but need not give the same subdivisions. Overflow or an unresolved nonfinite bound is failure, and underflow must be rounded outwards. The explicit Taylor rules and correctly rounded elementary evaluations are alternative enclosing realizations; the choice used for the reported lower-certificate statistics is specified below.

For general elementary evaluations one can use the Taylor polynomial with its remainder, without relying on a library of special functions. For $|z| \leq R$, the error after degree n in $\exp z$ is at most $e^R R^{n+1}/(n+1)!$; range reduction and squaring give rational upper bounds for the factor e^R . Sine and cosine are its real and imaginary parts. The primitive $\int_0^z e^{-t^2/2} dt$ is entire and is obtained by integrating the same power series. Square roots are enclosed by integer comparisons on the positive axis, and by the standard real-imaginary formula on rectangles avoiding the branch cut. Refining a rectangle is allowed when a branch or sign is unresolved.

A.2 Exact lower-certificate rules

The fiber-cover statistics in the proof of Theorem 3.5 refer to the binary realization of Appendix A.1 with 180 significand bits and root grid $2^{-80}\mathbb{Z}$. Rational parameters, polynomial coefficients, box vertices, sign tests, quadratic penalties and matrix comparisons are computed exactly before conversion to intervals. Square roots, exponentials, erf and π use correctly rounded lower and upper evaluations. In particular, the Gaussian primitive is evaluated as $E(x) = \sqrt{\pi/2} \text{erf}(x/\sqrt{2})$ by these interval operations, and cubic pieces use the cancellation-preserving formula below. The 220/96

replay means 220 significant bits and root grid $2^{-96}\mathbb{Z}$ on the same covering trees. The fixed-grid and Taylor–Mills rules below give an alternative realization with $p = 180$ and $U = 2^{180}$; regenerating a cover with it need not reproduce the reported tree statistics. All fiber parameters are defined by the integer vectors and exact index rules in Section 3.5, including the cost corrections and additional rows. All matrices use the same reference parameters α_0, κ_0 . The following rules specify the vertex enclosures used in Eq. (17).

Normalizer and phase constants. Write

$$A_n(q) = \sum_{j=0}^n \frac{(-1)^j q^{2j+1}}{2j+1}, \quad \pi_- = 16A_{41}(1/5) - 4A_{40}(1/239), \quad \pi_+ = 16A_{40}(1/5) - 4A_{41}(1/239).$$

Machin’s identity and the alternating series give $\pi_- < \pi < \pi_+$. Use $\text{out}_{180}([\pi_-, \pi_+])$ and its positive square root to enclose $S_0 = \sqrt{\pi}/2$. Put

$$C_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j}}{(2j)!}, \quad S_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{(2j+1)!},$$

$$c_- = C_9(\pi_+/7), \quad c_+ = C_{10}(\pi_-/7), \quad d_- = S_9(\pi_-/14), \quad d_+ = S_{10}(\pi_+/14).$$

Alternation and monotonicity enclose $\cos(\pi/7)$ by $[c_-, c_+]$ and $\sin(\pi/14)$ by $[d_-, d_+]$. The phase checks are the finite rational inequalities

$$0 < d_- < d_+ < c_- < c_+ < 1, \quad \frac{2\pi_+}{7(c_-^3 - d_+^3)} < a_0, \quad c_+^2 + c_+d_+ + d_+^2 < 1/\rho,$$

$$\frac{c_-^5 - d_+^5}{c_+^3 - d_-^3} > \sigma, \quad 2a_0 < c_0\pi_-, \quad S_0 < 251/200.$$

The last inequality is tested with the upper endpoint of its enclosure. The seven-sign inequality has already been proved analytically in Lemma 3.1; it requires no search.

Exponential and Gaussian primitive. For a grid point $x = u/U \geq 0$, enclose e^{-x} as follows. If $x \geq 180$, use $[0, U^{-1}]$, since $e > 2$. Otherwise choose $k = \max(0, \lfloor \log_2 u \rfloor + 1 - 180)$ for $u > 0$, and $k = 0$ for $u = 0$. Then $y = x/2^k \in [0, 1]$. With interval arithmetic form $t_0 = 1$, $t_j = t_{j-1}y/j$, and $P_n = \sum_{j=0}^n (-1)^j t_j$ through $n = 64$. Use $[\max(0, P_{63}^-), P_{64}^+]$ and square it k times. Here superscripts $-$ and $+$ denote lower and upper interval endpoints. For an interval of inputs use the endpoint evaluations in reversed order, because the exponential is decreasing.

To enclose $E(x) = \int_0^x e^{-z^2/2} dz$ at a nonnegative grid point, use $[\max(0, S_0^- - (e^{-x^2/2}/x)^+), S_0^+]$ if $x \geq 8$. Otherwise take $n = 40$ for $x \leq 2$, $n = 80$ for $2 < x \leq 4$, and $n = 160$ for $4 < x < 8$. Starting with $t_0 = x$ use

$$t_j = t_{j-1} \frac{(x^2/2)(2j-1)}{j(2j+1)}, \quad P_n = \sum_{j=0}^n (-1)^j t_j,$$

and return $[\max(0, P_{n-1}^-), \min(S_0^+, P_n^+)]$. For interval inputs evaluate its two endpoints in increasing order. These bounds follow by integrating the odd and even Taylor bounds for the exponential. They are valid even when the initial terms do not decrease. The rule for $x \geq 8$ is Mills’ inequality. Thus all primitive evaluations have proved errors.

Root covers and signs. Remove zero leading coefficients of a polynomial, and multiply its rational coefficients by a common positive denominator. Signs at $z = k2^{-80}$ are then signs of integers, obtained by homogeneous Horner evaluation. Every root interval below is rounded outwards to the root grid, intersected with $[0, 16]$, sorted, and merged with overlapping or touching intervals.

For a nonconstant polynomial P , define a cover recursively. A linear polynomial has its exact rational root enclosed on the grid. For higher degree, first cover all zeros of P' . Retain every interval of this cover, including those that need not contain a zero of P . The complementary intervals are intervals of monotonicity of P . Retain any endpoint where $P = 0$. When the two endpoint signs are opposite, bisect using exact signs until the root lies in one grid cell; an exact zero midpoint is retained as a singleton. Merge all retained intervals. Constants have empty covers. Induction on degree proves that every zero is covered, including multiple zeros: a multiple zero lies in the derivative cover.

For degree at most three a shorter equivalent derivative cover is used. Writing $P(z) = c_0 + c_1z + c_2z^2 + c_3z^3$, enclose the real numbers

$$\frac{-2c_2 \pm \sqrt{4c_2^2 - 12c_3c_1}}{6c_3}$$

when $c_3 \neq 0$ and the discriminant is nonnegative. If $c_3 = 0, c_2 \neq 0$, enclose $-c_1/(2c_2)$ instead; if both vanish use the empty cover. Square roots here use the outward arithmetic above before conversion to the root grid. These are all derivative roots, so the same monotonicity argument applies. For degree four or five use the recursive cover of P' .

Positive-part integration. For $P(z) = \sum_{j=0}^5 c_j z^j$, let $B_P = \sum_j |c_j| 16^j$. On each complementary monotone interval, equal nonnegative endpoint signs give a positive piece; equal nonpositive signs contribute zero. Opposite signs give one crossing isolated by the grid bisection above. Integrate the positive pieces using the primitive recurrence in Section 3.6. Every retained sliver $[l, r]$ contributes an interval $[0, (r - l)B_P]$. Add

$$[0, e^{-128} \sum_j |c_j| t_j], \quad t_0 = 1/16, \quad t_1 = 1, \quad t_j = 16^{j-1} + (j-1)t_{j-2}$$

for the tail above 16, and intersect the final enclosure with $[0, \infty)$. For cubic pieces the cancellation-preserving primitive formula is

$$c_0 \Delta E + c_1(e_l - e_r) + c_2(\Delta E + le_l - re_r) + c_3((l^2 + 2)e_l - (r^2 + 2)e_r), \quad \Delta E = E(r) - E(l).$$

For higher degree use $\sum_j c_j (J_j(r) - J_j(l))$.

The finite covering-tree test. For every row compute all derived parameters rationally. Check the positive denominators, the positive-definite matrices and the semidefinite comparisons by their diagonals and determinants. Check the inverse-coordinate and rank-one inverse-update identities, the effective exterior matrix and the scalar exterior inequality in Section 3.6. The oblique direction is included in both comparisons and in the clipping conjugate; it is not replaced by the zeroth-moment direction.

Start with the single box in Eq. (16). First apply the exact rational criterion of Lemma 3.8; accept a box satisfying it without computing any vertex integral. For every remaining box, at each rational vertex evaluate the four cubics in Eq. (12), add their positive-part integral enclosures, and add the exact rational clipping correction minus the complete quadratic penalty. In particular

retain the constant $-\kappa_0/\theta$ in every cubic and both cross terms in the penalty. The symmetries $(A, B) \mapsto (-A, -B)$ and $(M, N) \mapsto (-M, -N)$ justify the two nonnegative coordinates; on $A = 0$ or $M = 0$ they may also avoid duplicate vertex evaluations.

Use weights (e, q, o_a, o_c) for the variance bound. On a clipped row, evaluate $h_A A + h_B B$ at the four even corners of the box. Only if all four values are at least a , or all are at most $-a$, replace the first two weights by $e - h_A^2/d_0, q - h_B^2/d_0$. Check that these weights are positive. For an ordinary row make no replacement. With the selected weights, set

$$V_B = (w_A d_A^2 + w_B d_B^2 + o_a d_M^2 + o_c d_N^2)/4.$$

For a box not accepted by the zero-region criterion, the fixed-grid realization accepts exactly when

$$\max_v \mathcal{R}(v)^+ + \text{out}_{180}([V_B, V_B])^+ < 2^{-180} \lfloor 2^{180} I \rfloor.$$

For the reported binary realization, instead convert the exact rational V_B to a binary interval, add it to each vertex residual enclosure, and accept exactly when every resulting upper endpoint is strictly below $I = 0$. This is the same full-box sufficient inequality with a different outward-rounding rule. Otherwise use the midpoint, tie and depth rules in the text and process both children. Exhausting each tree proves that every point of the compact domain is in an accepted leaf. The convexity inequality or the zero-region lemma proves the required nonpositivity of the residual on each full leaf, and the exterior and symmetry arguments complete Theorem 3.5. The exact rational rule in Lemma 3.19 likewise regenerates the full scalar partition; neither box counts nor minimum margins substitute for these acceptance tests.

A.3 Exact upper-certificate rules

All elementary evaluations use outward interval arithmetic as in Section A.1. A terminating decimal is an exact rational input. Lower endpoints are rounded down and upper endpoints up. Division by a zero-containing interval, an unresolved nonfinite enclosure, or an incomplete cover is a failure. Correctly rounded elementary evaluations and Taylor expansions with proved remainders are interchangeable enclosing realizations. The parameters in Section 2 and the finite recurrences below specify the entire calculation; no additional numerical input is supplied. Intermediate arrays are generated by these recurrences, not assumed to satisfy the acceptance inequalities.

The finite Fourier and circle calculations. Generate the Bell polynomials from the recurrence of Lemma 2.3. Laurent polynomial addition, multiplication and differentiation are exact rational operations. For $0 \leq z \leq 1$, the sum through degree 80 of e^z is a lower bound and its remainder is at most

$$\frac{z^{81}}{81!} \frac{1}{1 - z/82}.$$

Repeated halving and squaring enclose larger positive arguments; reciprocation encloses negative arguments. Round interval endpoints outward to the grid 2^{-240} after every operation.

Work in $\mathbb{Q}[i][T, T^{-1}]$ and put

$$D(T) = A + \sum_{j=1}^3 \frac{jB_j}{4} (T^j + T^{-j}), \quad \partial(T^k) = \frac{ik}{2} T^k,$$

where A, B_1, B_2, B_3 are the exact coefficients in Eq. (3). Set $P_{0,0} = 1$, $P_{n,i} = 0$ for $i < 0$ or $i > n$, and $P_{n,0} = 0$ for $n > 0$. For $0 \leq n < 6$ form

$$P_{n+1,i} = \partial P_{n,i} + D P_{n,i-1} \quad (1 \leq i \leq n+1).$$

With $\mathcal{E}(T^k) = e^{-k^2/8}$, define

$$G_{n,s,t} = \mathcal{E}(P_{n,s}P_{n,t}), \quad \mu_{ij}^{(6)} = \sum_{l=0}^6 \binom{6}{l} \sum_{s=0}^i \sum_{t=0}^j G_{l,s,t} G_{6-l,i-s,j-t}.$$

Absent entries are zero. The two-variable Bell polynomial is the convolution of the two one-variable Bell polynomials, so independence of the coordinates proves this formula. Collect equal frequencies exactly before applying $\mathcal{E}$, and sum frequencies and the displayed indices in increasing order. Only $|k| \leq 36$ occurs. Opposite frequencies are conjugates, so their imaginary parts cancel exactly. These are whole-space Gaussian moments, with no spatial cutoff.

Form the six d_j by their formula in Section 2.1. Require $0 < d_j < \bar{d}_j$. The resulting enclosures are

j	strict lower bound for d_j	strict upper bound for d_j
0	0.24484499120775	0.24484499120776
1	6.68518338137632	6.68518338137634
2	296.7228852142933	296.7228852142934
3	4208.318469542131	4208.318469542132
4	27527.73432967899	27527.73432967901
5	70846.92623442754	70846.92623442756

Every upper endpoint is below the corresponding exact cap.

For the circle calculations, use the explicit Machin bounds π_-, π_+ from Appendix A.2. Put $J = 16384$ and enclose $\theta_j = j\pi/(2J)$, $0 \leq j \leq J$. Monotonicity gives the whole-arc rectangle

$$\mathcal{T}_j(s) = s([\cos \theta_{j+1}, \cos \theta_j] + i[\sin \theta_j, \sin \theta_{j+1}]).$$

Enclose all four endpoints, not just rounded angles. At zero and $\pi/2$ include the exact endpoint values explicitly. On the slightly extended endpoint intervals, $|\theta| < 8/5$; sine through degree 41 and cosine through degree 40 have error at most $[-2^{-110}, 2^{-110}]$. Starting with $p = r_{21}$, replace it by $p\mathcal{T}_j(s)^2 + r_k$ for $k = 19, 17, \dots, 1$, and finally by $p\mathcal{T}_j(s)$.

A binary64 interval operation uses the usual endpoint formula, with each computed lower endpoint replaced by its next representable predecessor and each upper endpoint by its successor. Products use all four endpoint products. Squaring uses lower bound zero when the interval contains zero. Square-root bounds l, h are accepted only after outward tests prove $l^2 \leq a^-$ and $h^2 \geq a^+$ for the input interval. Exact decimals may be enclosed by digit-by-digit integer accumulation followed by division by a power of ten. Complex arithmetic uses these real interval operations. The alternating Machin sums may themselves be evaluated in this arithmetic.

At radius $5/4$, require $|\Re \rho| < 39/40$ and $|\rho|^2 < (53/40)^2$ on every arc, and check the squared nuclear-norm cap in Lemma 2.4. At radius $107/100$, let X_j be the larger absolute real-coordinate endpoint, require $1 - X_j^2 > 0$, and form $z_j = (1 - X_j^2)^{-1}$. Evaluate $\sqrt{z_j} \sum_{l=0}^5 \bar{d}_l z_j^l$ by Horner's rule. Add its upper endpoints in increasing arc order, rounding up after each addition and after division by J . Require the result to be less than 28220000. Every arc must pass; no partial cover or nominated angle is accepted.

Conditional Hermite values. Put

$$q_0(z) = \operatorname{erf}(z/\sqrt{2}), \quad q_a(z) = \sqrt{2/\pi} e^{-z^2/2} h_{a-1}(-z)/\sqrt{a} \quad (a \geq 1).$$

Gaussian integration by parts gives $q_a(z) = \mathbb{E}[h_a(W) \operatorname{sgn}(W+z)]$. In particular $|q_a(z)| \leq 1$ for real z . Generate normalized Hermites by

$$h_0 = 1, \quad h_1 = x, \quad h_{j+1} = (xh_j - \sqrt{j}h_{j-1})/\sqrt{j+1}.$$

For conditional values use

$$e_0 = \sqrt{2/\pi} e^{-z^2/2}, \quad e_1 = -ze_0, \quad e_{j+1} = (-ze_j - \sqrt{j}e_{j-1})/\sqrt{j+1},$$

so $q_a = e_{a-1}/\sqrt{a}$ for $a \geq 1$. Differentiation gives, also for $a = 0$,

$$q_a^{(k)}(z) = \sqrt{\frac{(a+k-1)!}{a!}} e_{a+k-1}(z), \quad |q_a^{(k)}(z)| \leq \sqrt{\frac{(a+k)!}{a!}} \quad (k \geq 1).$$

For the last inequality, use $e_n = \sqrt{n+1} q_{n+1}$ and $|q_{n+1}| \leq 1$; thus it also covers $a = 0$ without a separate estimate.

The finite coefficient integrals. Put $N = 301$ and retain the complete domain

$$\mathcal{D} = \{(a, b, c) \in \mathbb{N}^3 : a + b + c \leq N, a + b + c \text{ odd}\}.$$

It has $M = 2329628$ elements. Integration in W gives

$$A_{a,b,c} = \int_{\mathbb{R}^2} q_a(u(x, y)) h_b(x) h_c(y) \varphi(x) \varphi(y) dx dy,$$

where φ is the standard Gaussian density. Use the tensor trapezoidal rule with spacing $h = 1/40$ and nodes $x_i = i/40$, $-480 \leq i \leq 480$. For every triple in $\mathcal{D}$ its analytic error satisfies

$$|A_{a,b,c} - h^2 \sum_{i,j=-480}^{480} q_a(u(x_i, x_j)) h_b(x_i) h_c(x_j) \varphi(x_i) \varphi(x_j)| < 10^{-12}. \quad (23)$$

Here is a uniform bound including both the infinite-grid error and the omitted nodes. On the polystrip $|\Im z_1|, |\Im z_2| \leq s = 7/10$, the exact coefficients of $v(x) = Ax + \sum_{j=1}^3 B_j \sin(jx/2)$ give

$$|\Im u(z_1, z_2)| \leq 2(|A|s + \sum_{j=1}^3 |B_j| \sinh(js/2)) < 63/10.$$

Complex Mehler with $\lambda = 13/20$ gives, for $a \geq 1$,

$$|q_a(\xi + i\zeta)| \leq (1 - \lambda^2)^{-1/4} \lambda^{-(a-1)/2} \exp\left(\frac{\zeta^2}{2(1 - \lambda)}\right).$$

Also $|q_0(\xi + i\zeta)| \leq 1 + |\zeta|e^{\zeta^2/2}$. A common bound for all $a \leq N$ is $Q < 10^{54}$; the enclosing comparison

$$3(20/13)^{151} e^{567/10} < 10^{54}$$

suffices for $a \geq 1$. Hermite translation and Cauchy–Schwarz give

$$\int_{\mathbb{R}} |h_b(x + is) \varphi(x + is)| dx \leq e^{s^2/2 + s\sqrt{b}}.$$

Indeed, the squared L^2 norm of the translated Hermite is at most $\sum_j (bs^2)^j / (j!)^2 \leq e^{2s\sqrt{b}}$. Since $b + c \leq 301$, every horizontal face or corner integral of the two-variable integrand is bounded by

$$M_{\text{strip}} < 10^{54} e^{s^2 + s\sqrt{602}} < 10^{62}.$$

Contour shifts and the two-dimensional Poisson formula bound the infinite-grid error by

$$M_{\text{strip}}\left(\frac{4}{E-1} + \frac{4}{(E-1)^2}\right) < 5 \cdot 10^{-14}, \quad E = e^{2\pi s/h} > 10^{76}.$$

The same estimates give Gaussian decay along the polystrip and justify the contour shifts and Poisson summation.

On the real axes use $|q_a| \leq 1$ and Mehler with $\lambda_0 = 99/100$. The absolute integrand is bounded by

$$8 \exp\left(-\frac{x^2 + y^2}{2(1 + \lambda_0)}\right).$$

For the one-variable Gaussian in this expression, the full weighted grid sum is at most $h + \sqrt{2\pi(1 + \lambda_0)} < 4$. The weighted sum outside $[-12, 12]$ is at most

$$\frac{2(1 + \lambda_0)}{12} e^{-144/(2(1 + \lambda_0))} < 10^{-15}$$

by the monotone integral test and Mills' inequality. The omitted two-dimensional sum is therefore less than $8 \cdot 2 \cdot 4 \cdot 10^{-15}$. Together these bounds prove Eq. (23).

Point evaluations and arithmetic errors. Here and below fl denotes rounding to IEEE binary64, to nearest with ties to even and gradual underflow. Each displayed addition and multiplication is rounded separately: contraction into a fused operation, reassociation, and flushing subnormal numbers to zero are excluded. An output NaN or infinity is failure.

First generate the input tables by directed interval arithmetic with 1024 significand bits. Let $\mathcal{I}(f)$ be the enclosure obtained from the displayed expression for f , enclosing every operation and elementary function. To store f , take $\widehat{f} = \text{fl}(\mathcal{I}(f)^-)$ and check the claimed error against *both* endpoints of $\mathcal{I}(f)$. In particular, an enclosure narrower than the tolerance does not by itself certify the rounding error. For the nodes, generate $x_i = i/40$, evaluate the four terms of v in their displayed order, and use the normalized Hermite recurrence to form $W_{b,i} = h\varphi(x_i)h_b(x_i)$ for $0 \leq b \leq 301$. These interval tests give

$$|\widehat{v}_i - v(x_i)| < 10^{-13}, \quad |\widehat{W}_{b,i} - h\varphi(x_i)h_b(x_i)| < 10^{-15}.$$

Every exact and every stored weight row has ℓ_1 norm below 2, checked by enclosing its complete finite sum.

For completeness, the Taylor table is generated at $z_j = j/32$, $0 \leq j \leq 640$. At each center generate $e_0, \dots, e_{312}$ by the recurrence above. For $0 \leq a \leq 301$ put

$$d_{a,1} = 1, \quad d_{a,k} = d_{a,k-1} \sqrt{a+k-1}/k \quad (2 \leq k \leq 12),$$

$$c_{a,j,0} = q_a(z_j), \quad c_{a,j,k} = d_{a,k} e_{a+k-1}(z_j) \quad (1 \leq k \leq 12).$$

Thus $c_{a,j,k} = q_a^{(k)}(z_j)/k!$. Store these values by the preceding rounding rule and require $|\widehat{c}_{a,j,k} - c_{a,j,k}| < 10^{-14} 18^k/k!$ for every index. The elementary primitives used here are specified in Section A.1; for example, a direct rational alternative for q_0 is

$$q_0(z) = \sqrt{2/\pi} \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{2^n n! (2n+1)}.$$

For $0 \leq z \leq 20$, even and odd partial sums enclose this value by integrating the corresponding Taylor bounds for $e^{-z^2/2}$. Increasing the truncation order and precision until the stated coefficient tests pass gives an alternative enclosing implementation.

For $|z| \leq 20$, use the nearest nonnegative center $j/32$ to $|z|$ and a degree-12 Taylor polynomial for q_a ; parity supplies negative arguments. Each stored Taylor coefficient has absolute error less than $10^{-14}18^k/k!$. Since $a + k \leq 314 < 18^2$, the derivative bound above gives $|q_a^{(k)}| \leq 18^k$ for $k \leq 13$. The Taylor remainder is at most

$$\frac{(18/64)^{13}}{13!} < 10^{-15}.$$

For stored arguments with absolute value greater than 20, the exact argument has absolute value greater than 19. Real Mehler and Mills' inequality justify replacing q_0 by the sign and q_a , $a \geq 1$, by zero; the error is below $20e^{-361/4} < 10^{-30}$.

The precise evaluation order is as follows. For a pair of nodes put $z = \text{fl}(\widehat{v}_i + \widehat{v}_j)$ and $z_0 = |z|$. If $z_0 \leq 20$, choose $j_0 = \lfloor 32z_0 + 1/2 \rfloor$ (an exact integer comparison), and set $d = z_0 - j_0/32$. This subtraction is exact in binary64 and $|d| \leq 1/64$. Start with $q = \widehat{c}_{a,j_0,12}$ and, for $k = 11, 10, \dots, 0$, replace it by

$$q \leftarrow \text{fl}(\widehat{c}_{a,j_0,k} + \text{fl}(dq)).$$

If $z_0 > 20$, set $q = 1$ for $a = 0$ and $q = 0$ otherwise. In either case, negate q precisely when $z < 0$ and a is even, using $q_a(-z) = (-1)^{a+1}q_a(z)$. The result is $\widehat{Q}_{a,i,j}$. Check $|\widehat{v}_i| < 32$ and $|\widehat{Q}_{a,i,j}| \leq 1.0000000001$ throughout.

Let $\epsilon_f = 2^{-53}$ and $\gamma_j = j\epsilon_f/(1 - j\epsilon_f)$. The sum of two stored threshold values differs from its exact value by at most $2 \cdot 10^{-13} + 64\epsilon_f$. Taylor-coefficient rounding, Horner evaluation and the remainder above then give

$$|\widehat{q}_a - q_a(u(x_i, x_j))| < 10^{-11}.$$

For clarity, the exact error test uses

$$S_{12} = \sum_{k=0}^{12} \frac{(18/64)^k}{k!} < 2$$

and checks

$$18(2 \cdot 10^{-13} + 64\epsilon_f) + 10^{-14}S_{12} + \gamma_{24}S_{12}(1 + 10^{-14}) + \frac{(18/64)^{13}}{13!} + 10^{-250} < 10^{-11}.$$

The small additive term accounts for possible underflow.

Define an ordered dot product by $s_{-481} = 0$ and

$$s_i = \text{fl}(s_{i-1} + \text{fl}(p_i q_i)) \quad (-480 \leq i \leq 480), \quad \text{dot}(p, q) = s_{480}.$$

For every $a = 0, \dots, 301$, form

$$\widehat{P}_{a,b,j} = \text{dot}_i(\widehat{W}_{b,i}, \widehat{Q}_{a,i,j}), \quad \widehat{A}_{a,b,c} = \text{dot}_j(\widehat{P}_{a,b,j}, \widehat{W}_{c,j}).$$

Retain precisely the triples in $\mathcal{D}$; each is used once, including both (a, b, c) and (a, c, b) when $b \neq c$. In particular there is no symmetry division or unprinted multiplicity. Set $d_w = 961 \cdot 10^{-15}$ and $g = \gamma_{1922}$. The row norm bounds and the standard dot-product error estimate, together with Eq. (23), give

$$|\widehat{A}_{a,b,c} - A_{a,b,c}| \leq 10^{-12} + 4d_w + 2d_w 10^{-11} + 4 \cdot 10^{-11} + 4g(1 + 10^{-11})(2 + g) + 10^{-250} < 10^{-10}.$$

All numerical comparisons in this paragraph are exact rational tests or enclosing elementary evaluations. The middle computation uses IEEE binary64 round-to-nearest with gradual underflow; its operation counts and magnitude bounds make the displayed underflow allowances more than sufficient.

The complete-array norm check. In increasing lexicographic order $(a, b, c) \in \mathcal{D}$, accumulate

$$S_0 = 0, \quad S_{j+1} = \text{up}(\text{fl}(S_j + |\hat{A}_j|)),$$

where up is the next representable successor. Require the exact count $M = 2329628$ and $S_M < 46$. Each step rounds upward from the exact sum of its binary inputs, so this proves the complete-array hypothesis

$$\sum_{(a,b,c) \in \mathcal{D}} |\hat{A}_{a,b,c}| < 46.$$

A bound for a partial array does not satisfy this test.

Ordered coefficient assembly. Set $\hat{r}_j = \text{fl}(r_j)$ for $0 \leq j \leq 21$, taking every unlisted coefficient to be zero, and $\hat{\pi} = \text{fl}((\pi_- + \pi_+)/2)$. Check, by exact rational comparisons against the input enclosures,

$$|\hat{r}_j - r_j| < 10^{-16}, \quad |\hat{\pi} - \pi| < 10^{-15}.$$

Initialize all $\hat{C}_{a,n}$ to zero. For $a = 0, \dots, 301$ and $0 \leq n \leq 301 - a$ with $a + n$ odd, start with $c = 0$ and, for $b = 0, 1, \dots, n$, perform

$$c \leftarrow \text{fl}(c + \text{fl}(\hat{A}_{a,b,n-b}^2)).$$

Then set $\hat{C}_{a,n} = (-1)^n c$. All entries with $a + n$ even remain zero. Independently count the retained triples, requiring

$$\#\mathcal{D} = \sum_{m=1,3,\dots,301} \binom{m+2}{2} = 2329628.$$

The following truncated Horner recurrence specifies every final coefficient. Initialize $f_0, \dots, f_{301}$ to zero. For $a = 301, 300, \dots, 0$, initialize a new zero vector g . For each $n = 0, \dots, 301 - a$, start with $v = \hat{C}_{a,n}$ and, in increasing order $j = 1, \dots, \min(21, n)$, replace

$$v \leftarrow \text{fl}(v + \text{fl}(\hat{r}_j f_{n-j})).$$

Set $g_n = v$ and, after finishing this stage, replace the whole vector f by g . Finally set $\hat{b}_n = \text{fl}(f_n(\hat{\pi}/2))$. All omitted higher degrees are irrelevant: ρ has zero constant term, so an entry omitted at stage a cannot contribute to a final degree at most 301. Check $\hat{b}_n = 0$ at every even index.

Interpret each finite binary number $\hat{b}_n$ as its exact dyadic rational value. Compute

$$\hat{B} = \hat{b}_1 - \sum_{m=3,5,\dots,301} |\hat{b}_m|$$

with integer numerator and denominator arithmetic, *not* an additional floating-point sum. Require $\hat{B} > 0.882216067$. All input-table checks, all arcs, all moment caps, the complete triple count, the complete-array norm check, and this final comparison must pass in the same calculation.

The complete-array norm converts the individual integration errors into an aggregate coefficient error.

Lemma A.1 (Aggregate integration error). *For the complete finite coefficient vector computed by these rules, with all point, row-norm, count and complete-array tests passed,*

$$\sum_{m \leq 301} |b_m - \hat{b}_m| < 0.000000016.$$

Proof. Put $\varepsilon = 10^{-10}$. The uniform integral error and the complete-array test give

$$\sum_{\mathcal{D}} |\hat{A}_{a,b,c}^2 - A_{a,b,c}^2| \leq 2\varepsilon \sum_{\mathcal{D}} |\hat{A}_{a,b,c}| + M\varepsilon^2 < 92\varepsilon + M\varepsilon^2.$$

Every power of the exact reservoir polynomial has absolute coefficient sum at most one. Thus the map from squared Hermite coefficients to the coefficient vector of H has ℓ_1 norm at most $\pi/2$.

The remaining arithmetic forms rounded versions of

$$C_{a,n} = (-1)^n \sum_{b+c=n} \hat{A}_{a,b,c}^2.$$

It then evaluates $\sum_a \rho(t)^a \sum_n C_{a,n} t^n$ by polynomial Horner steps, truncated through degree 301. Positive square sums use γ_{604} ; reservoir input error is at most $\delta = 22 \cdot 10^{-16}$; all Horner paths use $\gamma_{44.302}$. Parseval still gives $\sum_{\mathcal{D}} A_{a,b,c}^2 \leq 1$, so the computed square sum is at most $(1 + \sqrt{M}\varepsilon)^2$. In detail, put

$$V = (1 + 1527\varepsilon)^2, \quad V_1 = (1 + \gamma_{604})V, \quad I = (1 + \delta)^{302}.$$

The total additional assembly error, including the rounded $\pi/2$, is at most

$$E_{\text{ass}} := 2\{\gamma_{604}V + (I - 1)V_1 + \gamma_{44.302}IV_1\} + 2 \cdot 10^{-15}IV_1 + 10^{-250} < 10^{-9}.$$

This is an exact rational comparison. Keeping the displayed value of E_{ass} , the full error is bounded by

$$\frac{\pi+}{2}(92 \cdot 10^{-10} + 2329628 \cdot 10^{-20}) + E_{\text{ass}} < 1.6 \cdot 10^{-8}.$$

This final comparison is rational. In particular the sharper error estimate explicitly requires the complete-array cap 46. $\square$

The final comparisons use the exact dyadic interpretation just specified; rounded decimal displays do not replace it. The complete tail uses the rational quantities in Lemma 2.5, with no truncated exterior or extrapolated coefficients.

Aggregate bounds. We now apply the finite verification rules to certify the coefficient head and the six moment caps.

Lemma A.2 (Aggregate certificate). *The quantities defined in Section 2 satisfy*

$$b_1 - \sum_{3 \leq m \leq 301, m \text{ odd}} |b_m| > \ell_* := 0.88221605, \quad 0 < d_j < \bar{d}_j \quad (0 \leq j \leq 5).$$

Proof. Apply the finite recurrences and acceptance tests of Appendix A.3. The Fourier calculation gives all six displayed moment enclosures. The circle calculation covers all 16384 quarter-circle arcs at each radius. Directed point tables and ordered products then compute all 2329628 required Hermite integrals. The following ledger records enclosing bounds checked in the same calculation:

quantity	verified upper bound
$\max_{ t =5/4} \Re \rho(t) $	0.971031784
$\max_{ t =5/4} \rho(t) $	1.321639198
$\max_{ t =107/100} \Re \rho(t) $	0.842762088
mean sixth-order nuclear majorant	28209773.802
exact and stored weight-row ℓ_1 norms	1.000000000001
$\sum_{\mathcal{D}} \hat{A}_{a,b,c} $	45.916634669 < 46
$\sum_{m=3,5,\dots,301} \hat{b}_m $	0.00000040914750
$\max_{\mathcal{D}} A_{a,b,c} - \hat{A}_{a,b,c} $	10^{-10}
$\sum_{m=0}^{301} b_m - \hat{b}_m $	0.000000016
$\sum_{m \geq 303, m \text{ odd}} b_m $	0.000000085

The first seven rows are finite interval or exact rational tests of the printed recurrences; the last three are the proved error and tail bounds. The complete-array accumulation gives the exact dyadic value

$$S_M = \frac{1615547959235395}{35184372088832} < 46.$$

Interpreting every binary output coefficient as an exact dyadic rational also gives

$$\hat{b}_1 - \sum_{3 \leq m \leq 301, m \text{ odd}} |\hat{b}_m| = \frac{18762660725857602231430074482130989295}{2^{124}} > 0.882216067.$$

The ledger and these identities are outputs of the finite generating rules, not additional inputs. Each final strict comparison is an integer comparison after clearing positive denominators. Lemma A.1 gives

$$b_1 - \sum_{3 \leq m \leq 301, m \text{ odd}} |b_m| > 0.882216067 - 0.000000016 = 0.882216051 > \ell_*.$$

Together with Lemma 2.5, this covers every coefficient. The remaining exact comparison is

$$0.882216051 - 0.000000085 - \frac{1102769}{1250000} = 0.000000001 > 0.$$

Lemma 2.1 gives $K_G \leq 625000\pi/1102769 < 1.78051379$. □